

# Service Parts Catalog - Parte 3

HP LaserJet Series

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TechGuide IA — Consulta Inteligente de Manuais HP

a. Release the support shaft (callout 1), and then slide it toward the center of the door (callout 2). TIP: It might be easier to release the support shaft by using a small flat-blade screwdriver. Figure 5-695 Release the support shaft 1 2 b. Remove the support shaft. Figure 5-696 Remove the support shaft Removal and replacement: Fan (FM1) 865

c. Release one tab (callout 1), and then slide the guide out of the printer (callout 2). CAUTION: Be careful, the tab (callout 1) can be easily broken. Figure 5-697 Remove the guide 2 1 d. Use a small flat-blade screwdriver to gently pry the white-plastic door arm (callout 1) off of the gray-plastic pressure-release link (callout 2) to release the door arm. Figure 5-698 Release the pressure-release link 1 2 866 Chapter 5 Removal and replacement

e. With the door in the partially closed position, slightly raise the left end (callout 1; raise it over the sheet-metal hinge boss), and then slide the door to the left (callout 2) to release the right hinge (callout 3). CAUTION: The door is still attached to the printer by the left-side retainer arm. Do not attempt to completely remove the door. Reinstallation tip: When reinstalling the door, there are two right-side hinge pins (not shown in callout 3). Keep a finger pressed on the left-side retainer arm so that it does not disengage from the door. Figure 5-699 Release the door 3 1 2

f. Rotate the door away from the printer until the keyed pin on the left-side retainer arm can disengage from the door. Figure 5-700 Release the left-side retainer arm Removal and replacement: Fan (FM1) 867

g. Remove the door. Figure 5-701 Remove the door 3. Remove the rear door assembly. a. At the rear of the printer, remove two screws (callout 1). Figure 5-702 Remove two screws 1 868 Chapter 5 Removal and replacement

b. Rotate the bottom edge of the assembly away from the printer to release it. Figure 5-703 Rotate the bottom of the cover c. Pull down on the assembly to remove it. Figure 5-704 Remove the cover 4. Remove the left cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. NOTE: An MFP printer is shown in this section. However, the procedure is correct for all printer models. Removal and replacement: Fan (FM1) 869

a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-705 Locate of the tabs and bosses b. Remove one screw (callout 1). Figure 5-706 Remove one screw 1 870 Chapter 5 Removal and replacement

c. At the front of the printer, release one boss (callout 1), and then rotate the front edge of the cover away from the printer to release three tabs (callout 2). Figure 5-707 Release one boss and three tabs 2 1 d. At the bottom edge of the cover, release two tabs. Figure 5-708 Release two tabs e. Continue to rotate the front edge of the cover away from the printer (callout 1), and then slide the cover toward the rear of the printer (callout 2) to remove it. NOTE: If the cover is difficult to rotate or slide as shown below, make sure that the bottom tabs (released in the previous step) are completely disengaged from the chassis. Removal and replacement: Fan (FM1) 871

Figure 5-709 Remove the cover 1 2 5. Remove the fan (FM1). a. At the left side of the printer, disconnect one connector (callout 1), and then release the wire harness from the retainer (callout 2). Reinstallation tip: It might be easier to connect the connector first, and then install the wire harness in the retainer. Figure 5-710 Disconnect one connector 1 2 872 Chapter 5 Removal and replacement

b. Release one tab (callout 1), and then remove the fan duct (callout 2). Figure 5-711 Remove the fan duct 1 2 c. Remove one screw (callout 1), and then remove the film bias assembly (callout 2). Figure 5-712 Remove the film bias assembly 1 2 d. Remove the fan (callout 2). CAUTION: When reinstalled the fan must draw air into the printer. Make sure that the two arrows embossed on the top of the fan body (near the wire harness) face upward and into the printer. Removal and replacement: Fan (FM1) 873

Figure 5-713 Remove the fan 6. Unpack the replacement assembly. a. Dispose of the defective part. NOTE: HP recommends responsible disposal of the defective part. <http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> b. Unpack the replacement part from the packaging. CAUTION: Some parts are sensitive to electrostatic discharge (ESD). Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. If the replacement part is a roller or pad, avoid touching the spongy part of it. Skin oils on the roller or pad can cause paper pickup and/or print-quality problems. HP recommends using disposable gloves when handling rollers or pads or washing your hands before touching rollers or pads. IMPORTANT: Make sure that all of the shipping materials (for example shipping tape) are removed from the replacement part prior to installation. c. To install an assembly, reverse the removal steps. NOTE: When applicable, special installation instructions are provided for an assembly at the end of the removal procedure. Always completely read the removal instructions and follow all special installation instructions. Removal and replacement: Fan (FM2) Learn how to remove and replace the fan (FM2). To view a short video of this procedure, click the link below. 874 Chapter 5 Removal and replacement

M501 View a video of how to remove and replace the fan (FM2) (M501).

## M506/E50045

View a video of how to remove and replace the fan (FM2) (M506/E50045).

## M507/E50145

View a video of how to remove and replace the fan (FM2) (M507/E50145).

## M527/E52545/M528/E52645

View a video of how to remove and replace the fan (FM2) (M527/E52545/M528/E52645). Mean time to repair: 12 minutes Service level: Easy

## Before performing service

### Turn the printer power off

■ Disconnect the power cable. **WARNING!** To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. Use the table below to identify the correct part number for your printer. To order the part, go to [www.hp.com/buy/parts](http://www.hp.com/buy/parts). Table 5-38 Part information

### Part number Part description

RK2-8068-000CN Fan (FM2)

### Required tools

● Small flat-blade screwdriver ● #2 Phillips screwdriver with a magnetic tip and a 152 mm (6 in) shaft length

### After performing service

### Turn the printer power on

■ Connect the power cable. ■ Use the power switch to turn the power on.

### Post service test

No post service test is available for this assembly. 1. Remove the control panel (M506x/E50045x/M507x/E50145x). Removal and replacement: Fan (FM2) 875

a. Tilt the control panel to the fully raised position. Figure 5-714 Tilt the control panel up b. Use the small flat-blade screwdriver supplied in the kit to release the control-panel screws cover, and then remove the cover. **TIP:** The screwdriver has a reversible shaft. Simply pull the shaft out of the handle, and then reinstall it with the desired end facing out. Figure 5-715 Remove the cover 876 Chapter 5 Removal and replacement

c. Use the small Phillips screwdriver supplied in the kit to remove two screws (callout 1). **IMPORTANT:** Store the screws in a secure place where they cannot fall down into the printer. Figure 5-716 Remove two screws 1 d. Turn the control panel over (callout 1), to gain access to the bottom side. Figure 5-717 Turn the control panel over 1 Removal and replacement: Fan (FM2) 877

e. M506/E50045 only: Disconnect one HDMI cable (callout 1), one connector (callout 2), and then remove the control panel. Figure 5-718 Remove the control panel (M506/E50045) 1 2 f. M507/E50154 only: If the formatter cover has not been removed, remove it now. Disconnect one flat-flexible cable (FFC). **CAUTION:** ESD sensitive part. **NOTE:** This connector is a zero-insertion force (ZIF) connector. Open the locking clip. The clip is easily dislodged and lost. Figure 5-719 Disconnect one FFC (M507/E50145) 878 Chapter 5 Removal and replacement

g. M507/E50145 only: Pass the FFC through the opening in the cover, and then remove the control panel. **CAUTION:** ESD sensitive part. Figure 5-720 Remove the control panel (M507/E50145) 2. Remove the tray. a. Pull the tray straight out of the printer until it stops. Figure 5-721 Pull the tray out until it stops Removal and replacement: Fan (FM2) 879

b. Lift the front of the tray, and then pull it out of the printer to remove it. **NOTE:** If the tray has Fixed Tray Guides installed and are still required, move the hardware to new tray. If necessary, order new kit: 3GY19-67901 Kit-Fixed Tray Guide. Only one kit required per engine. Figure 5-722 Release and remove the tray 3. Remove the formatter cover. ■ Do one of the following: ● M501: Remove one screw (callout 1), and then slide the cover toward the rear of the printer to remove it (callout 2). Figure 5-723 Remove the formatter cover (M501) 2 1 ● M506/E50045/M507/E50145/M527/E52545/M528/E52645: Slide the cover toward the rear of the printer to remove it. 880 Chapter 5 Removal and replacement

Figure 5-724 Remove the formatter cover (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

4. Remove the hard-disk drive (HDD models). **CAUTION:** ESD sensitive part. a. Release the locking connector (callout 1), and then pinch the retainer (callout 2) to release it. Figure 5-725 Release the HDD 1 2 Removal and replacement: Fan (FM2) 881

b. Rotate the connector end of the HDD out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-726 Remove the HDD 1 2 5. Remove the fax PCA (M527/E52545/M528/E52645 fax models). **CAUTION:** ESD-sensitive part. a. Disconnect three connectors (callout 1). Figure 5-727 Disconnect three connectors 1 882 Chapter 5 Removal and replacement

b. Pinch the retainer to release it. Figure 5-728 Release the fax PCA c. Rotate the connector end of the fax PCA out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-729 Remove the fax PCA 1 2 Removal and replacement: Fan (FM2) 883

6. Remove the formatter. **CAUTION:** Never remove and then install a replacement DCC and formatter at the same time. Doing so will cause the printer to become unstable. If a replacement DCC and formatter must be installed, first remove and install the formatter. Test operate the printer, and then if necessary install a replacement DCC. **IMPORTANT:** Take note of all empty connectors on the PCA before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessory items. a. M501: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. Figure 5-730 Remove the formatter (M501) 1 b.

M506/E50045/M507/E50145/M527/E52545/M528/E52645: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. **IMPORTANT:** Take note of all empty connectors on the formatter before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessories items. 884 Chapter 5 Removal and replacement

Figure 5-731 Remove the formatter (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

1 1 7. Remove the stapler door or stapler blank cover (M527/E52545/M528/E52645). a. Do one of the following: ● Stapler models: Open the stapler door. ● Non-stapler models: Use a small flat-blade screw driver to release two tabs (callout 1), and then rotate the cover away from the printer (callout 2). Figure 5-732 Open the door or release the cover 2 1 b. Do one of the following: ● Stapler models: Use a small flat-blade screwdriver to release the upper hinge. ● Non-stapler models: Slide the cover toward the front of the printer to remove it. Removal and replacement: Fan (FM2) 885

Figure 5-733 Release the upper hinge or remove the cover c. Stapler models: Use a small flat-blade screwdriver to release the lower hinge.

Figure 5-734 Release the lower hinge 886 Chapter 5 Removal and replacement

d. Stapler models: Remove the door. Figure 5-735 Remove the door 8. Remove the stapler stationary (inner) cover (M527/E52545/M528/E52645). **NOTE:** Stapler models only. a. Before proceeding, take note of the locations of the upper tab (callout 1) and the lower tab (callout 2) on the stapler stationary (inner) cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs. Figure 5-736 Locate the tabs 1 2 Removal and replacement: Fan (FM2) 887

b. Use a long-shaft small flat-blade screwdriver to release the lower tab. Figure 5-737 Release one tab c. Use a long-shaft small flat-blade screwdriver to release the upper tab. Figure 5-738 Release one tab 888 Chapter 5 Removal and replacement

d. With the tabs disengaged the front of the cover is released. Rotate the front of the cover away from the printer (callout 1) to release the rear of the cover (callout 2), and then remove it. Figure 5-739 Remove the cover 1 2 9. Remove the top-left cover (M527/E52545/M528/E52645). a. Open the toner-cartridge door, release two tabs (callout 1), and then rotate the rear edge of the cover away from the printer. Figure 5-740 Release two tabs 1 Removal and replacement: Fan (FM2) 889

b. Continue to rotate the cover away from the printer to release two more tabs at the middle of the cover. Figure 5-741 Release two tabs c. Slide the cover toward the rear of the printer to release two bosses at the front of the cover. Figure 5-742 Slide the cover 890 Chapter 5 Removal and replacement

d. Remove the cover. Figure 5-743 Remove the cover 10. Remove the top-rear cover (M527/E52545/M528/E52645). a. Remove one screw (callout 1), and then open the rear door. Figure 5-744 Remove one screw 1 Removal and replacement: Fan (FM2) 891

b. Rotate the exposed edge of the cover away from the printer (as shown). Figure 5-745 Rotate the cover c. Continue to rotate the cover away from the printer to release one boss. Figure 5-746 Release one boss 892 Chapter 5 Removal and replacement

d. Remove the cover. Figure 5-747 Remove the cover 11. Remove the rear door assembly. a. At the rear of the printer, remove two screws (callout 1). Figure 5-748 Remove two screws 1 Removal and replacement: Fan (FM2) 893

b. Rotate the bottom edge of the assembly away from the printer to release it. Figure 5-749 Rotate the bottom of the cover c. Pull down on the assembly to remove it. Figure 5-750 Remove the cover 12. Remove the cartridge door assembly. **NOTE:** If the toner cartridge was not removed prior to servicing the printer, remove it now. a. Release the support shaft (callout 1), and then slide it toward the center of the door (callout 2). **TIP:** It might be easier to release the support shaft by using a small flat-blade screwdriver. 894 Chapter 5 Removal and replacement

Figure 5-751 Release the support shaft 1 2 b. Remove the support shaft. Figure 5-752 Remove the support shaft c. Release one tab (callout 1), and then slide the guide out of the printer (callout 2). **CAUTION:** Be careful, the tab (callout 1) can be easily broken. Removal and replacement: Fan (FM2) 895

Figure 5-753 Remove the guide 2 1 d. Use a small flat-blade screwdriver to gently pry the white-plastic door arm (callout 1) off of the gray-plastic pressure-release link (callout 2) to release the door arm. Figure 5-754 Release the pressure-release link 1 2 e. With the door in the partially closed position, slightly raise the left end (callout 1; raise it over the sheet-metal hinge boss), and then slide the door to the left (callout 2) to release the right hinge (callout 3). **CAUTION:** The door is still attached to the printer by the left-side retainer arm. Do not attempt to completely remove the door. Reinstallation tip: When reinstalling the door, there are two right-side hinge pins (not shown in callout 3). Keep a finger pressed on the left-side retainer arm so that it does not disengage from the door. 896 Chapter 5 Removal and replacement

Figure 5-755 Release the door 3 1 2 f. Rotate the door away from the printer until the keyed pin on the left-side retainer arm can disengage from the door. Figure 5-756 Release the left-side retainer arm Removal and replacement: Fan (FM2) 897

g. Remove the door. Figure 5-757 Remove the door 13. Remove the right cover. **IMPORTANT:** The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-758 Locate the tabs and bosses 898 Chapter 5 Removal and replacement

b. Remove two screws (callout 1). Figure 5-759 Remove two screws 1 c. At the rear of the printer, release one boss. **CAUTION:** It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-760 Release one boss d. At the bottom edge of the cover, release two bosses. **CAUTION:** It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Removal and replacement: Fan (FM2) 899

Figure 5-761 Release two bosses e. Near the formatter, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Hold the bottom of the cover away from the printer, and then use a small flat-blade screw driver to easily release this tab. Figure 5-762 Release one tab 900 Chapter 5 Removal and replacement

f. At the top rear edge of the cover, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Use a small flat-blade screw driver to easily release this tab. Figure 5-763 Release one tab g. Rotate the rear of the cover away from the printer (callout 1), slide the cover toward the front of the printer (callout 2) to release two bosses (callout 3), and then remove the cover. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-764 Remove the cover 2 1 3 Removal and replacement: Fan (FM2) 901

14. Remove the left cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. NOTE: An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-765 Locate of the tabs and bosses b. Remove one screw (callout 1). Figure 5-766 Remove one screw 1 902 Chapter 5 Removal and replacement

c. At the front of the printer, release one boss (callout 1), and then rotate the front edge of the cover away from the printer to release three tabs (callout 2). Figure 5-767 Release one boss and three tabs 2 1 d. At the bottom edge of the cover, release two tabs. Figure 5-768 Release two tabs e. Continue to rotate the front edge of the cover away from the printer (callout 1), and then slide the cover toward the rear of the printer (callout 2) to remove it. NOTE: If the cover is difficult to rotate or slide as shown below, make sure that the bottom tabs (released in the previous step) are completely disengaged from the chassis. Removal and replacement: Fan (FM2) 903

Figure 5-769 Remove the cover 1 2 15. Remove the integrated-scanner assembly (ISA) (M527/E52545/M528/E52645). NOTE: The ISA includes the document feeder, sub-scanner assembly (SSA), and the scanner control board (SCB). a. Remove one screw (callout 1). IMPORTANT: This screw fastens the ISA ground plate to the printer chassis. Do not lose this screw. Figure 5-770 Remove one screw 1 904 Chapter 5 Removal and replacement

b. Disconnect three connectors (callout 1), pass the wire harnesses and cables through the opening in the cover (callout 2), and then release them from the retainers (callout 3). NOTE: One cable (callout 4) does not need to pass through the opening in the cover. TIP: It might be easier if the WiFi PCA and holder (not all models; callout 5) is detached from the printer, and then moved out of the way. Figure 5-771 Disconnect cables and connectors 4 1 2 3 5 c. Disconnect three connectors (callout 1), and then remove six screws (callout 2). Figure 5-772 Disconnect connectors and remove screws 1 2 Removal and replacement: Fan (FM2) 905

d. Slightly slide the ISA towards the back of the printer to release it. Figure 5-773 Slide the ISA towards the back of the printer e. Remove the ISA. Figure 5-774 Remove the ISA 16. Remove the top cover (M527/E52545/M528/E52645). 906 Chapter 5 Removal and replacement

a. At the front right of the printer, remove one screw (callout 1), and then remove the sheet-metal plate (callout 2). Figure 5-775 Remove one screw and the plate 1 2 b. At the top rear of the printer, remove three screws (callout 1). Figure 5-776 Remove three screws 1 c. At the top front of the printer, remove three screws (callout 1). NOTE: The screw (callout 2) on the lower-right portion of the cover is a different color than the other two screws, and is surrounded by sheet-metal. It might appear that this screw does not need to be removed. Remove this screw. Removal and replacement: Fan (FM2) 907

Figure 5-777 Remove three screws 1 2 d. At the front left of the printer, release one boss. Figure 5-778 Release one boss e. Rotate the front of the cover up and away from the printer. CAUTION: As the cover is rotated up, carefully pass the cables and wire harnesses the hole in the cover (callout 1). 908 Chapter 5 Removal and replacement

Figure 5-779 Rotate the front of the cover up 1 f. Remove the cover. Figure 5-780 Remove the cover 17. Remove the top cover (M501/M506/E50045/M507/E50145). The M506x/E50045x is shown in this procedure. However, the steps are correct for all M506/E50045/M507/E50145 models and the M501 printer. Removal and replacement: Fan (FM2) 909

a. Remove two screws (callout 1). Figure 5-781 Remove two screws 1 b. M506/E50045/ only: Release one cable retainer from the chassis. TIP: Depress the tab on the retainer to release it. Figure 5-782 Release one cable retainer 910 Chapter 5 Removal and replacement

c. M507/E50145 only: Disconnect two connectors. Figure 5-783 Disconnect two connectors d. Release one boss. Figure 5-784 Release one boss e. M506/E50045/ only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M506x/E50045x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. Removal and replacement: Fan (FM2) 911

Figure 5-785 Remove the cover (M506/E50045) 1 f. M507/E50145only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M507x/E50145x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). IMPORTANT: If the top cover is being replaced, make sure to transfer the WiFi PCA from the discarded cover to the replacement cover. NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. Figure 5-786 Remove the cover (M507/E50145) 1 912 Chapter 5 Removal and replacement

18. Remove the paper delivery tray (output bin). NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. a. Release two tabs (callout 1), and then slide the cover toward the front of the printer (callout 2). Figure 5-787 Release two tabs 1 2 1 b. Remove the bin. Figure 5-788 Remove the bin 19. Remove the formatter case. Removal and replacement: Fan (FM2) 913

a. M506/E50045/M507/E50145 only: Remove two screws (callout 1), and then remove the sheet metal plate (callout 2). NOTE: M507/E50145 only: The M507/E50145 sheet-metal plate includes a separate black plastic cable guide (not shown). Do not lose the guide when removing the plate. Figure 5-789 Remove the sheet-metal plate 2 1 b. Carefully pass all cables and wire harnesses through the opening in the formatter case. NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. Figure 5-790 Pass all cables and wire harnesses through the opening 914 Chapter 5 Removal and replacement

c. Remove one screw (callout 1), and then push down on one tab (callout 2) to release it. Figure 5-791 Remove one screw 1 2 d. Remove the guide. Figure 5-792 Remove the guide Removal and replacement: Fan (FM2) 915

e. Remove four screws (callout 1). Figure 5-793 Remove four screws 1 f. Remove five screws (callout 1). Figure 5-794 Remove five screws 1 916 Chapter 5 Removal and replacement

g. Disconnect one connector (callout 1), and then move the guide out of the way (callout 2). Figure 5-795 Disconnect one connector 2 1 h. Slightly lift up the formatter case (callout 1), rotate the top of the case away from the chassis (callout 2), and then lift up the case to release it (callout 3). CAUTION: Pass the FFC through the opening (callout 4) in the formatter while removing the case. Figure 5-796 Release the formatter case 2 3 1 4 i. Remove the formatter case. NOTE: WiFi models only: If the formatter case is being replaced, make sure that the WiFi PCA and bracket (callout 1) are transferred to the replacement case. Removal and replacement: Fan (FM2) 917

Figure 5-797 Remove the formatter case 1 20. Remove the fan (FM2). a. Disconnect one connector (callout 1), and then release the wire harness from the retainer (callout 2). Figure 5-798 Disconnect one connector 1 2 918 Chapter 5 Removal and replacement

b. Release two tabs (callout 1), and then rotate the top of the fan out and away from the printer (callout 2). Figure 5-799 Release two tabs 2 1

c. Remove the fan. CAUTION: When reinstalled the fan must draw air into the printer. Make sure that the two arrows embossed on the right side of the fan body face downward and into the printer. Figure 5-800 Remove the fan 21. Unpack the replacement assembly. a. Dispose of the defective part. NOTE: HP recommends responsible disposal of the defective part.

<http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> Removal and replacement: Fan (FM2) 919

b. Unpack the replacement part from the packaging. CAUTION: Some parts are sensitive to electrostatic discharge (ESD). Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. If the replacement part is a roller or pad, avoid touching the spongy part of it. Skin oils on the roller or pad can cause paper pickup and/or print-quality problems. HP recommends using disposable gloves when handling rollers or pads or washing your hands before touching rollers or pads. IMPORTANT: Make sure that all of the shipping materials (for example shipping tape) are removed from the replacement part prior to installation. c. To install an assembly, reverse the removal steps. NOTE: When applicable, special installation instructions are provided for an assembly at the end of the removal procedure. Always completely read the removal instructions and follow all special installation instructions. Removal and replacement: Fuser power supply (FPS) Learn how to remove and replace the fuser power supply (FPS). To view a short video of this procedure, click the link below. M501 View a video of how to remove and replace the FPS (M501).

## M506/E50045

View a video of how to remove and replace the FPS (M506/E50045).

## M507/E50145

View a video of how to remove and replace the FPS (M507/E50145).

## M527/E52545/M528/E52645

View a video of how to remove and replace the FPS (M527/E52545/M528/E52645). Mean time to repair: 12 minutes Service level: Easy

### Before performing service

#### Turn the printer power off

■ Disconnect the power cable. WARNING! To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. 920 Chapter 5 Removal and replacement

Use the table below to identify the correct part number for your printer. To order the part, go to [www.hp.com/buy/parts](http://www.hp.com/buy/parts). Table 5-39 Part information

| Part number    | Part description                             |
|----------------|----------------------------------------------|
| RM2-7948-000CN | Fixing (fuser) power supply (FPS; 110V-127V) |
| RM2-7949-000CN | Fixing (fuser) power supply (FPS; 220V-240V) |

#### Required tools

- Small flat-blade screwdriver
- #2 Phillips screwdriver with a magnetic tip and a 152 mm (6 in) shaft length

### After performing service

#### Turn the printer power on

- Connect the power cable.
- Use the power switch to turn the power on.

## Post service test

Make sure that the printer initializes to a Ready state. 1. Remove the control panel (M506x/E50045x/M507x/E50145x). a. Tilt the control panel to the fully raised position. Figure 5-801 Tilt the control panel up Removal and replacement: Fuser power supply (FPS) 921

b. Use the small flat-blade screwdriver supplied in the kit to release the control-panel screws cover, and then remove the cover. TIP: The screwdriver has a reversible shaft. Simply pull the shaft out of the handle, and then reinstall it with the desired end facing out. Figure 5-802 Remove the cover c. Use the small Phillips screwdriver supplied in the kit to remove two screws (callout 1). IMPORTANT: Store the screws in a secure place where they cannot fall down into the printer. Figure 5-803 Remove two screws 1 922 Chapter 5 Removal and replacement

d. Turn the control panel over (callout 1), to gain access to the bottom side. Figure 5-804 Turn the control panel over 1 e. M506/E50045 only: Disconnect one HDMI cable (callout 1), one connector (callout 2), and then remove the control panel. Figure 5-805 Remove the control panel (M506/E50045) 1 2 f. M507/E50154 only: If the formatter cover has not been removed, remove it now. Disconnect one flat-flexible cable (FFC). CAUTION: ESD sensitive part. NOTE: This connector is a zero-insertion force (ZIF) connector. Open the locking clip. The clip is easily dislodged and lost. Removal and replacement: Fuser power supply (FPS) 923

Figure 5-806 Disconnect one FFC (M507/E50145) g. M507/E50145 only: Pass the FFC through the opening in the cover, and then remove the control panel. CAUTION: ESD sensitive part. Figure 5-807 Remove the control panel (M507/E50145) 2. Remove the tray. 924 Chapter 5 Removal and replacement

a. Pull the tray straight out of the printer until it stops. Figure 5-808 Pull the tray out until it stops b. Lift the front of the tray, and then pull it out of the printer to remove it. NOTE: If the tray has Fixed Tray Guides installed and are still required, move the hardware to new tray. If necessary, order new kit: 3GY19-67901 Kit-Fixed Tray Guide. Only one kit required per engine. Figure 5-809 Release and remove the tray 3. Remove the formatter cover. ■ Do one of the following: ● M501: Remove one screw (callout 1), and then slide the cover toward the rear of the printer to remove it (callout 2). Removal and replacement: Fuser power supply (FPS) 925

Figure 5-810 Remove the formatter cover (M501) 2 1 ● M506/E50045/M507/E50145/M527/E52545/M528/E52645: Slide the cover toward the rear of the printer to remove it. Figure 5-811 Remove the formatter cover (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

4. Remove the hard-disk drive (HDD models). CAUTION: ESD sensitive part. 926 Chapter 5 Removal and replacement

a. Release the locking connector (callout 1), and then pinch the retainer (callout 2) to release it. Figure 5-812 Release the HDD 1 2 b. Rotate the connector end of the HDD out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-813 Remove the HDD 1 2 5. Remove the fax PCA (M527/E52545/M528/E52645 fax models). CAUTION: ESD-sensitive part. Removal and replacement: Fuser power supply (FPS) 927

a. Disconnect three connectors (callout 1). Figure 5-814 Disconnect three connectors 1 b. Pinch the retainer to release it. Figure 5-815 Release the fax PCA 928 Chapter 5 Removal and replacement

c. Rotate the connector end of the fax PCA out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-816 Remove the fax PCA 1 2 6. Remove the formatter. CAUTION: Never remove and then install a replacement DCC and formatter at the same time. Doing so will cause the printer to become unstable. If a replacement DCC and formatter must be installed, first remove and install the formatter. Test operate the printer, and then if necessary install a replacement DCC. IMPORTANT: Take note of all empty connectors on the PCA before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessory items. Removal and replacement: Fuser power supply (FPS) 929

a. M501: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. Figure 5-817 Remove the formatter (M501) 1 b. M506/E50045/M507/E50145/M527/E52545/M528/E52645: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. IMPORTANT: Take note of all empty connectors on the formatter before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessories items. Figure 5-818 Remove the formatter (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

1 1 7. Remove the stapler door or stapler blank cover (M527/E52545/M528/E52645). a. Do one of the following: ● Stapler models: Open the stapler door. ● Non-stapler models: Use a small flat-blade screw driver to release two tabs (callout 1), and then rotate the cover away from the printer (callout 2). 930 Chapter 5 Removal and replacement

Figure 5-819 Open the door or release the cover 2 1 b. Do one of the following: ● Stapler models: Use a small flat-blade screwdriver to release the upper hinge. ● Non-stapler models: Slide the cover toward the front of the printer to remove it. Figure 5-820 Release the upper hinge or remove the cover Removal and replacement: Fuser power supply (FPS) 931

c. Stapler models: Use a small flat-blade screwdriver to release the lower hinge. Figure 5-821 Release the lower hinge d. Stapler models: Remove the door. Figure 5-822 Remove the door 8. Remove the stapler stationary (inner) cover (M527/E52545/M528/E52645). NOTE: Stapler models only. a. Before proceeding, take note of the locations of the upper tab (callout 1) and the lower tab (callout 2) on the stapler stationary (inner) cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs. 932 Chapter 5 Removal and replacement



Figure 5-823 Locate the tabs 1 2 b. Use a long-shaft small flat-blade screwdriver to release the lower tab. Figure 5-824 Release one tab Removal and replacement: Fuser power supply (FPS) 933

c. Use a long-shaft small flat-blade screwdriver to release the upper tab. Figure 5-825 Release one tab d. With the tabs disengaged the front of the cover is released. Rotate the front of the cover away from the printer (callout 1) to release the rear of the cover (callout 2), and then remove it. Figure 5-826 Remove the cover 1 2 9. Remove the top-left cover (M527/E52545/M528/E52645). 934 Chapter 5 Removal and replacement

a. Open the toner-cartridge door, release two tabs (callout 1), and then rotate the rear edge of the cover away from the printer. Figure 5-827 Release two tabs 1 b. Continue to rotate the cover away from the printer to release two more tabs at the middle of the cover. Figure 5-828 Release two tabs Removal and replacement: Fuser power supply (FPS) 935

c. Slide the cover toward the rear of the printer to release two bosses at the front of the cover. Figure 5-829 Slide the cover d. Remove the cover. Figure 5-830 Remove the cover 10. Remove the top-rear cover (M527/E52545/M528/E52645). 936 Chapter 5 Removal and replacement

a. Remove one screw (callout 1), and then open the rear door. Figure 5-831 Remove one screw 1 b. Rotate the exposed edge of the cover away from the printer (as shown). Figure 5-832 Rotate the cover Removal and replacement: Fuser power supply (FPS) 937

c. Continue to rotate the cover away from the printer to release one boss. Figure 5-833 Release one boss d. Remove the cover. Figure 5-834 Remove the cover 11. Remove the rear door assembly. 938 Chapter 5 Removal and replacement

a. At the rear of the printer, remove two screws (callout 1). Figure 5-835 Remove two screws 1 b. Rotate the bottom edge of the assembly away from the printer to release it. Figure 5-836 Rotate the bottom of the cover Removal and replacement: Fuser power supply (FPS) 939

c. Pull down on the assembly to remove it. Figure 5-837 Remove the cover 12. Remove the cartridge door assembly. NOTE: If the toner cartridge was not removed prior to servicing the printer, remove it now. a. Release the support shaft (callout 1), and then slide it toward the center of the door (callout 2). TIP: It might be easier to release the support shaft by using a small flat-blade screwdriver. 940 Chapter 5 Removal and replacement

Figure 5-838 Release the support shaft 1 2 b. Remove the support shaft. Figure 5-839 Remove the support shaft c. Release one tab (callout 1), and then slide the guide out of the printer (callout 2). CAUTION: Be careful, the tab (callout 1) can be easily broken. Removal and replacement: Fuser power supply (FPS) 941

Figure 5-840 Remove the guide 2 1 d. Use a small flat-blade screwdriver to gently pry the white-plastic door arm (callout 1) off of the gray-plastic pressure-release link (callout 2) to release the door arm. Figure 5-841 Release the pressure-release link 1 2 e. With the door in the partially closed position, slightly raise the left end (callout 1; raise it over the sheet-metal hinge boss), and then slide the door to the left (callout 2) to release the right hinge (callout 3). CAUTION: The door is still attached to the printer by the left-side retainer arm. Do not attempt to completely remove the door. Reinstallation tip: When reinstalling the door, there are two right-side hinge pins (not shown in callout 3). Keep a finger pressed on the left-side retainer arm so that it does not disengage from the door. 942 Chapter 5 Removal and replacement

Figure 5-842 Release the door 3 1 2 f. Rotate the door away from the printer until the keyed pin on the left-side retainer arm can disengage from the door. Figure 5-843 Release the left-side retainer arm Removal and replacement: Fuser power supply (FPS) 943

g. Remove the door. Figure 5-844 Remove the door 13. Remove the right cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-845 Locate the tabs and bosses 944 Chapter 5 Removal and replacement

b. Remove two screws (callout 1). Figure 5-846 Remove two screws 1 c. At the rear of the printer, release one boss. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-847 Release one boss d. At the bottom edge of the cover, release two bosses. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Removal and replacement: Fuser power supply (FPS) 945

Figure 5-848 Release two bosses e. Near the formatter, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Hold the bottom of the cover away from the printer, and then use a small flat-blade screw driver to easily release this tab. Figure 5-849 Release one tab 946 Chapter 5 Removal and replacement

f. At the top rear edge of the cover, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Use a small flat-blade screw driver to easily release this tab. Figure 5-850 Release one tab g. Rotate the rear of the cover away from the printer (callout 1), slide the cover toward the front of the printer (callout 2) to release two bosses (callout 3), and then remove the cover. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-851 Remove the cover 2 1 3 Removal and replacement: Fuser power supply (FPS) 947

14. Remove the left cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. NOTE: An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-852 Locate of the tabs and bosses b. Remove one screw (callout 1). Figure 5-853 Remove one screw 1 948 Chapter 5 Removal and replacement

c. At the front of the printer, release one boss (callout 1), and then rotate the front edge of the cover away from the printer to release three tabs (callout 2). Figure 5-854 Release one boss and three tabs 2 1 d. At the bottom edge of the cover, release two tabs. Figure 5-855 Release two tabs e. Continue to rotate the front edge of the cover away from the printer (callout 1), and then slide the cover toward the rear of the printer



(callout 2) to remove it. NOTE: If the cover is difficult to rotate or slide as shown below, make sure that the bottom tabs (released in the previous step) are completely disengaged from the chassis. Removal and replacement: Fuser power supply (FPS) 949

Figure 5-856 Remove the cover 1 2 15. Remove the integrated-scanner assembly (ISA) (M527/E52545/M528/E52645). NOTE: The ISA includes the document feeder, sub-scanner assembly (SSA), and the scanner control board (SCB). a. Remove one screw (callout 1).

IMPORTANT: This screw fastens the ISA ground plate to the printer chassis. Do not lose this screw. Figure 5-857 Remove one screw 1 950 Chapter 5 Removal and replacement

b. Disconnect three connectors (callout 1), pass the wire harnesses and cables through the opening in the cover (callout 2), and then release them from the retainers (callout 3). NOTE: One cable (callout 4) does not need to pass through the opening in the cover. TIP: It might be easier if the WiFi PCA and holder (not all models; callout 5) is detached from the printer, and then moved out of the way. Figure 5-858 Disconnect cables and connectors 4 1 2 3 5 c. Disconnect three connectors (callout 1), and then remove six screws (callout 2). Figure 5-859 Disconnect connectors and remove screws 1 2 Removal and replacement: Fuser power supply (FPS) 951

d. Slightly slide the ISA towards the back of the printer to release it. Figure 5-860 Slide the ISA towards the back of the printer e. Remove the ISA. Figure 5-861 Remove the ISA 16. Remove the top cover (M527/E52545/M528/E52645). 952 Chapter 5 Removal and replacement

a. At the front right of the printer, remove one screw (callout 1), and then remove the sheet-metal plate (callout 2). Figure 5-862 Remove one screw and the plate 1 2 b. At the top rear of the printer, remove three screws (callout 1). Figure 5-863 Remove three screws 1 c. At the top front of the printer, remove three screws (callout 1). NOTE: The screw (callout 2) on the lower-right portion of the cover is a different color than the other two screws, and is surrounded by sheet-metal. It might appear that this screw does not need to be removed. Remove this screw. Removal and replacement: Fuser power supply (FPS) 953

Figure 5-864 Remove three screws 1 2 d. At the front left of the printer, release one boss. Figure 5-865 Release one boss e. Rotate the front of the cover up and away from the printer. CAUTION: As the cover is rotated up, carefully pass the cables and wire harnesses the hole in the cover (callout 1). 954 Chapter 5 Removal and replacement

Figure 5-866 Rotate the front of the cover up 1 f. Remove the cover. Figure 5-867 Remove the cover 17. Remove the top cover (M501/M506/E50045/M507/E50145). The M506x/E50045x is shown in this procedure. However, the steps are correct for all M506/E50045/M507/E50145 models and the M501 printer. Removal and replacement: Fuser power supply (FPS) 955

a. Remove two screws (callout 1). Figure 5-868 Remove two screws 1 b. M506/E50045/ only: Release one cable retainer from the chassis. TIP: Depress the tab on the retainer to release it. Figure 5-869 Release one cable retainer 956 Chapter 5 Removal and replacement

c. M507/E50145 only: Disconnect two connectors. Figure 5-870 Disconnect two connectors d. Release one boss. Figure 5-871 Release one boss e. M506/E50045/ only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M506x/E50045x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. Removal and replacement: Fuser power supply (FPS) 957

Figure 5-872 Remove the cover (M506/E50045) 1 f. M507/E50145only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M507x/E50145x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). IMPORTANT: If the top cover is being replaced, make sure to transfer the WiFi PCA from the discarded cover to the replacement cover. NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. Figure 5-873 Remove the cover (M507/E50145) 1 958 Chapter 5 Removal and replacement

18. Remove the paper delivery tray (output bin). NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. a. Release two tabs (callout 1), and then slide the cover toward the front of the printer (callout 2). Figure 5-874 Release two tabs 1 2 1 b. Remove the bin. Figure 5-875 Remove the bin 19. Remove the formatter case. Removal and replacement: Fuser power supply (FPS) 959

a. M506/E50045/M507/E50145 only: Remove two screws (callout 1), and then remove the sheet metal plate (callout 2). NOTE: M507/E50145 only: The M507/E50145 sheet-metal plate includes a separate black plastic cable guide (not shown). Do not lose the guide when removing the plate. Figure 5-876 Remove the sheet-metal plate 2 1 b. Carefully pass all cables and wire harnesses through the opening in the formatter case. NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. Figure 5-877 Pass all cables and wire harnesses through the opening 960 Chapter 5 Removal and replacement

c. Remove one screw (callout 1), and then push down on one tab (callout 2) to release it. Figure 5-878 Remove one screw 1 2 d. Remove the guide. Figure 5-879 Remove the guide Removal and replacement: Fuser power supply (FPS) 961

e. Remove four screws (callout 1). Figure 5-880 Remove four screws 1 f. Remove five screws (callout 1). Figure 5-881 Remove five screws 1 962 Chapter 5 Removal and replacement

g. Disconnect one connector (callout 1), and then move the guide out of the way (callout 2). Figure 5-882 Disconnect one connector 2 1 h. Slightly lift up the formatter case (callout 1), rotate the top of the case away from the chassis (callout 2), and then lift up the case to release it (callout 3). CAUTION: Pass the FFC through the opening (callout 4) in the formatter while removing the case. Figure 5-883 Release the formatter case 2 3 1 4 i. Remove the formatter case. NOTE: WiFi models only: If the formatter case is being replaced, make sure that the WiFi PCA and bracket (callout 1) are transferred to the replacement case. Removal and replacement: Fuser power supply (FPS) 963

Figure 5-884 Remove the formatter case 1 20. Remove the fuser power supply (FPS). TIP: The FPS includes the optional paper feeder connector (Jetlink accessory tray connector) . a. Disconnect three connectors (callout 1). Figure 5-885 Disconnect three connectors 1 964 Chapter 5 Removal and replacement

b. Remove one screw (callout 1). Figure 5-886 Remove one screw 1 c. Disconnect the tray accessory connector (circle callout) and release the wire harness from the guide. d. Release four tabs. TIP: It might be easier to release the upper two tabs by using a small flat-blade screw driver. Removal and replacement: Fuser power supply (FPS) 965

Figure 5-887 Release four tabs e. Rotate the left end of the assembly out and away from the printer (callout 1), slide the assembly to the left to release the right side alignment pins (callout 2), and then remove the assembly. Figure 5-888 Remove the assembly 1 2 21. Unpack the

replacement assembly. a. Dispose of the defective part. NOTE: HP recommends responsible disposal of the defective part.

<http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> 966 Chapter 5 Removal and replacement

b. Unpack the replacement part from the packaging. CAUTION: Some parts are sensitive to electrostatic discharge (ESD). Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. If the replacement part is a roller or pad, avoid touching the spongy part of it. Skin oils on the roller or pad can cause paper pickup and/or print-quality problems. HP recommends using disposable gloves when handling rollers or pads or washing your hands before touching rollers or pads. IMPORTANT: Make sure that all of the shipping materials (for example shipping tape) are removed from the replacement part prior to installation. c. To install an assembly, reverse the removal steps. NOTE: When applicable, special installation instructions are provided for an assembly at the end of the removal procedure. Always completely read the removal instructions and follow all special installation instructions. Removal and replacement: DC controller Learn how to remove and replace the DC controller. CAUTION: Never remove and then install a replacement DCC and formatter at the same time. Doing so will cause the printer to become unstable. If a replacement DCC and formatter must be installed, first remove and install the formatter. Test operate the printer, and then if necessary install a replacement DCC. IMPORTANT: Some figure might show assemblies removed or installed that have not yet been removed at that specific step. However, the procedures are correct for this printer and the target assembly. Always thoroughly read the instructions that accompany each figure. To view a short video of this procedure, click the link below. M501 View a video of how to remove and replace the DCC (M501).

## M506/E50045

View a video of how to remove and replace the DCC (M506M506/E5004).

## M507/E50145

View a video of how to remove and replace the DCC (M507/E50145).

## M527/E52545/M528/E52645

View a video of how to remove and replace the DCC (M527/E52545/M528/E52645). Mean time to repair: 12 minutes Service level: Easy Removal and replacement: DC controller 967

### Before performing service

#### Turn the printer power off

■ Disconnect the power cable. WARNING! To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. Use the table below to identify the correct part number for your printer. To order the part, go to [www.hp.com/buy/parts](http://www.hp.com/buy/parts). Table 5-40 Part information

#### Part number Part description

RM2-8614-000CN M506/E50045 For simplex models - DC controller PC board (PCA) RM3-7476-000CN M507/E50145 For simplex models - DC controller PC board (PCA) RM2-8711-030CN M501dn/ M501n - DC controller PC board (PCA) RM2-8612-000CN M506/M527/E52545/E50045 For duplex models - DC controller PC board (PCA) RM3-7475-010CN M507/M528/E52645/E50145 For duplex models - DC controller PC board (PCA)

#### Required tools

● Small flat-blade screwdriver ● #2 Phillips screwdriver with a magnetic tip and a 152 mm (6 in) shaft length

### After performing service

#### Turn the printer power on

■ Connect the power cable. ■ Use the power switch to turn the power on.

#### Post service test

Make sure that the printer initializes to a Ready state. Print a configuration page to make sure that the printer is functioning correctly. 1. Remove the control panel (M506x/E50045x/M507x/E50145x). 968 Chapter 5 Removal and replacement

a. Tilt the control panel to the fully raised position. Figure 5-889 Tilt the control panel up b. Use the small flat-blade screwdriver supplied in the kit to release the control-panel screws cover, and then remove the cover. TIP: The screwdriver has a reversible shaft. Simply pull the shaft out of the handle, and then reinstall it with the desired end facing out. Figure 5-890 Remove the cover Removal and replacement: DC controller 969

c. Use the small Phillips screwdriver supplied in the kit to remove two screws (callout 1). **IMPORTANT:** Store the screws in a secure place where they cannot fall down into the printer. Figure 5-891 Remove two screws 1 d. Turn the control panel over (callout 1), to gain access to the bottom side. Figure 5-892 Turn the control panel over 1 970 Chapter 5 Removal and replacement

e. M506/E50045 only: Disconnect one HDMI cable (callout 1), one connector (callout 2), and then remove the control panel. Figure 5-893 Remove the control panel (M506/E50045) 1 2 f. M507/E50154 only: If the formatter cover has not been removed, remove it now. Disconnect one flat-flexible cable (FFC). **CAUTION:** ESD sensitive part. **NOTE:** This connector is a zero-insertion force (ZIF) connector. Open the locking clip. The clip is easily dislodged and lost. Figure 5-894 Disconnect one FFC (M507/E50145) Removal and replacement: DC controller 971

g. M507/E50145 only: Pass the FFC through the opening in the cover, and then remove the control panel. **CAUTION:** ESD sensitive part. Figure 5-895 Remove the control panel (M507/E50145) 2. Remove the tray. a. Pull the tray straight out of the printer until it stops. Figure 5-896 Pull the tray out until it stops 972 Chapter 5 Removal and replacement

b. Lift the front of the tray, and then pull it out of the printer to remove it. **NOTE:** If the tray has Fixed Tray Guides installed and are still required, move the hardware to new tray. If necessary, order new kit: 3GY19-67901 Kit-Fixed Tray Guide. Only one kit required per engine. Figure 5-897 Release and remove the tray 3. Remove the formatter cover. ■ Do one of the following: ● M501: Remove one screw (callout 1), and then slide the cover toward the rear of the printer to remove it (callout 2). Figure 5-898 Remove the formatter cover (M501) 2 1 ● M506/E50045/M507/E50145/M527/E52545/M528/E52645: Slide the cover toward the rear of the printer to remove it. Removal and replacement: DC controller 973

Figure 5-899 Remove the formatter cover (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

4. Remove the hard-disk drive (HDD models). **CAUTION:** ESD sensitive part. a. Release the locking connector (callout 1), and then pinch the retainer (callout 2) to release it. Figure 5-900 Release the HDD 1 2 974 Chapter 5 Removal and replacement

b. Rotate the connector end of the HDD out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-901 Remove the HDD 1 2 5. Remove the fax PCA (M527/E52545/M528/E52645 fax models). **CAUTION:** ESD-sensitive part. a. Disconnect three connectors (callout 1). Figure 5-902 Disconnect three connectors 1 Removal and replacement: DC controller 975

b. Pinch the retainer to release it. Figure 5-903 Release the fax PCA c. Rotate the connector end of the fax PCA out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-904 Remove the fax PCA 1 2 976 Chapter 5 Removal and replacement

6. Remove the formatter. **CAUTION:** Never remove and then install a replacement DCC and formatter at the same time. Doing so will cause the printer to become unstable. If a replacement DCC and formatter must be installed, first remove and install the formatter. Test operate the printer, and then if necessary install a replacement DCC. **IMPORTANT:** Take note of all empty connectors on the PCA before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessory items. a. M501: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. Figure 5-905 Remove the formatter (M501) 1 b. M506/E50045/M507/E50145/M527/E52545/M528/E52645: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. **IMPORTANT:** Take note of all empty connectors on the formatter before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessories items. Removal and replacement: DC controller 977

Figure 5-906 Remove the formatter (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

1 1 7. Remove the stapler door or stapler blank cover (M527/E52545/M528/E52645). a. Do one of the following: ● Stapler models: Open the stapler door. ● Non-stapler models: Use a small flat-blade screw driver to release two tabs (callout 1), and then rotate the cover away from the printer (callout 2). Figure 5-907 Open the door or release the cover 2 1 b. Do one of the following: ● Stapler models: Use a small flat-blade screwdriver to release the upper hinge. ● Non-stapler models: Slide the cover toward the front of the printer to remove it. 978 Chapter 5 Removal and replacement

Figure 5-908 Release the upper hinge or remove the cover c. Stapler models: Use a small flat-blade screwdriver to release the lower hinge. Figure 5-909 Release the lower hinge Removal and replacement: DC controller 979

d. Stapler models: Remove the door. Figure 5-910 Remove the door 8. Remove the stapler stationary (inner) cover (M527/E52545/M528/E52645). **NOTE:** Stapler models only. a. Before proceeding, take note of the locations of the upper tab (callout 1) and the lower tab (callout 2) on the stapler stationary (inner) cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs. Figure 5-911 Locate the tabs 1 2 980 Chapter 5 Removal and replacement

b. Use a long-shaft small flat-blade screwdriver to release the lower tab. Figure 5-912 Release one tab c. Use a long-shaft small flat-blade screwdriver to release the upper tab. Figure 5-913 Release one tab Removal and replacement: DC controller 981

d. With the tabs disengaged the front of the cover is released. Rotate the front of the cover away from the printer (callout 1) to release the rear of the cover (callout 2), and then remove it. Figure 5-914 Remove the cover 1 2 9. Remove the top-left cover (M527/E52545/M528/E52645). a. Open the toner-cartridge door, release two tabs (callout 1), and then rotate the rear edge of the cover away from the printer. Figure 5-915 Release two tabs 1 982 Chapter 5 Removal and replacement

b. Continue to rotate the cover away from the printer to release two more tabs at the middle of the cover. Figure 5-916 Release two tabs c. Slide the cover toward the rear of the printer to release two bosses at the front of the cover. Figure 5-917 Slide the cover Removal and replacement: DC controller 983

d. Remove the cover. Figure 5-918 Remove the cover 10. Remove the top-rear cover (M527/E52545/M528/E52645). a. Remove one screw (callout 1), and then open the rear door. Figure 5-919 Remove one screw 1 984 Chapter 5 Removal and replacement

b. Rotate the exposed edge of the cover away from the printer (as shown). Figure 5-920 Rotate the cover c. Continue to rotate the cover away from the printer to release one boss. Figure 5-921 Release one boss Removal and replacement: DC controller 985

d. Remove the cover. Figure 5-922 Remove the cover 11. Remove the rear door assembly. a. At the rear of the printer, remove two screws (callout 1). Figure 5-923 Remove two screws 1 986 Chapter 5 Removal and replacement

b. Rotate the bottom edge of the assembly away from the printer to release it. Figure 5-924 Rotate the bottom of the cover c. Pull down on the assembly to remove it. Figure 5-925 Remove the cover 12. Remove the cartridge door assembly. NOTE: If the toner cartridge was not removed prior to servicing the printer, remove it now. a. Release the support shaft (callout 1), and then slide it toward the center of the door (callout 2). TIP: It might be easier to release the support shaft by using a small flat-blade screwdriver. Removal and replacement: DC controller 987

Figure 5-926 Release the support shaft 1 2 b. Remove the support shaft. Figure 5-927 Remove the support shaft c. Release one tab (callout 1), and then slide the guide out of the printer (callout 2). CAUTION: Be careful, the tab (callout 1) can be easily broken. 988 Chapter 5 Removal and replacement

Figure 5-928 Remove the guide 2 1 d. Use a small flat-blade screwdriver to gently pry the white-plastic door arm (callout 1) off of the gray-plastic pressure-release link (callout 2) to release the door arm. Figure 5-929 Release the pressure-release link 1 2 e. With the door in the partially closed position, slightly raise the left end (callout 1; raise it over the sheet-metal hinge boss), and then slide the door to the left (callout 2) to release the right hinge (callout 3). CAUTION: The door is still attached to the printer by the left-side retainer arm. Do not attempt to completely remove the door. Reinstallation tip: When reinstalling the door, there are two right-side hinge pins (not shown in callout 3). Keep a finger pressed on the left-side retainer arm so that it does not disengage from the door. Removal and replacement: DC controller 989

Figure 5-930 Release the door 3 1 2 f. Rotate the door away from the printer until the keyed pin on the left-side retainer arm can disengage from the door. Figure 5-931 Release the left-side retainer arm 990 Chapter 5 Removal and replacement

g. Remove the door. Figure 5-932 Remove the door 13. Remove the right cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-933 Locate the tabs and bosses Removal and replacement: DC controller 991

b. Remove two screws (callout 1). Figure 5-934 Remove two screws 1 c. At the rear of the printer, release one boss. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-935 Release one boss d. At the bottom edge of the cover, release two bosses. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. 992 Chapter 5 Removal and replacement

Figure 5-936 Release two bosses e. Near the formatter, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Hold the bottom of the cover away from the printer, and then use a small flat-blade screw driver to easily release this tab. Figure 5-937 Release one tab Removal and replacement: DC controller 993

f. At the top rear edge of the cover, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Use a small flat-blade screw driver to easily release this tab. Figure 5-938 Release one tab g. Rotate the rear of the cover away from the printer (callout 1), slide the cover toward the front of the printer (callout 2) to release two bosses (callout 3), and then remove the cover. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-939 Remove the cover 2 1 3 994 Chapter 5 Removal and replacement

14. Remove the left cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. NOTE: An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-940 Locate of the tabs and bosses b. Remove one screw (callout 1). Figure 5-941 Remove one screw 1 Removal and replacement: DC controller 995

c. At the front of the printer, release one boss (callout 1), and then rotate the front edge of the cover away from the printer to release three tabs (callout 2). Figure 5-942 Release one boss and three tabs 2 1 d. At the bottom edge of the cover, release two tabs. Figure 5-943 Release two tabs e. Continue to rotate the front edge of the cover away from the printer (callout 1), and then slide the cover toward the rear of the printer (callout 2) to remove it. NOTE: If the cover is difficult to rotate or slide as shown below, make sure that the bottom tabs (released in the previous step) are completely disengaged from the chassis. 996 Chapter 5 Removal and replacement

Figure 5-944 Remove the cover 1 2 15. Remove the integrated-scanner assembly (ISA) (M527/E52545/M528/E52645). NOTE: The ISA includes the document feeder, sub-scanner assembly (SSA), and the scanner control board (SCB). a. Remove one screw (callout 1). IMPORTANT: This screw fastens the ISA ground plate to the printer chassis. Do not lose this screw. Figure 5-945 Remove one screw 1 Removal and replacement: DC controller 997

b. Disconnect three connectors (callout 1), pass the wire harnesses and cables through the opening in the cover (callout 2), and then release them from the retainers (callout 3). NOTE: One cable (callout 4) does not need to pass through the opening in the cover. TIP: It might be easier if the WiFi PCA and holder (not all models; callout 5) is detached from the printer, and then moved out of the way. Figure 5-946

Disconnect cables and connectors 4 1 2 3 5 c. Disconnect three connectors (callout 1), and then remove six screws (callout 2). Figure 5-947 Disconnect connectors and remove screws 1 2 998 Chapter 5 Removal and replacement

d. Slightly slide the ISA towards the back of the printer to release it. Figure 5-948 Slide the ISA towards the back of the printer e. Remove the ISA. Figure 5-949 Remove the ISA 16. Remove the top cover (M527/E52545/M528/E52645). Removal and replacement: DC controller 999

a. At the front right of the printer, remove one screw (callout 1), and then remove the sheet-metal plate (callout 2). Figure 5-950 Remove one screw and the plate 1 2 b. At the top rear of the printer, remove three screws (callout 1). Figure 5-951 Remove three screws 1 c. At the top front of the printer, remove three screws (callout 1). NOTE: The screw (callout 2) on the lower-right portion of the cover is a different color than the other two screws, and is surrounded by sheet-metal. It might appear that this screw does not need to be removed. Remove this screw. 1000 Chapter 5 Removal and replacement

Figure 5-952 Remove three screws 1 2 d. At the front left of the printer, release one boss. Figure 5-953 Release one boss e. Rotate the front of the cover up and away from the printer. CAUTION: As the cover is rotated up, carefully pass the cables and wire harnesses the hole in the cover (callout 1). Removal and replacement: DC controller 1001

Figure 5-954 Rotate the front of the cover up 1 f. Remove the cover. Figure 5-955 Remove the cover 17. Remove the top cover (M501/M506/E50045/M507/E50145). The M506x/E50045x is shown in this procedure. However, the steps are correct for all M506/E50045/M507/E50145 models and the M501 printer. 1002 Chapter 5 Removal and replacement

a. Remove two screws (callout 1). Figure 5-956 Remove two screws 1 b. M506/E50045/ only: Release one cable retainer from the chassis. TIP: Depress the tab on the retainer to release it. Figure 5-957 Release one cable retainer Removal and replacement: DC controller 1003

c. M507/E50145 only: Disconnect two connectors. Figure 5-958 Disconnect two connectors d. Release one boss. Figure 5-959 Release one boss e. M506/E50045/ only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M506x/E50045x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. 1004 Chapter 5 Removal and replacement

Figure 5-960 Remove the cover (M506/E50045) 1 f. M507/E50145only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M507x/E50145x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). IMPORTANT: If the top cover is being replaced, make sure to transfer the WiFi PCA from the discarded cover to the replacement cover. NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. Figure 5-961 Remove the cover (M507/E50145) 1 Removal and replacement: DC controller 1005

18. Remove the paper delivery tray (output bin). NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. a. Release two tabs (callout 1), and then slide the cover toward the front of the printer (callout 2). Figure 5-962 Release two tabs 1 2 1 b. Remove the bin. Figure 5-963 Remove the bin 19. Remove the formatter case. 1006 Chapter 5 Removal and replacement

a. M506/E50045/M507/E50145 only: Remove two screws (callout 1), and then remove the sheet metal plate (callout 2). NOTE: M507/E50145 only: The M507/E50145 sheet-metal plate includes a separate black plastic cable guide (not shown). Do not lose the guide when removing the plate. Figure 5-964 Remove the sheet-metal plate 2 1 b. Carefully pass all cables and wire harnesses through the opening in the formatter case. NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. Figure 5-965 Pass all cables and wire harnesses through the opening Removal and replacement: DC controller 1007

c. Remove one screw (callout 1), and then push down on one tab (callout 2) to release it. Figure 5-966 Remove one screw 1 2 d. Remove the guide. Figure 5-967 Remove the guide 1008 Chapter 5 Removal and replacement

e. Remove four screws (callout 1). Figure 5-968 Remove four screws 1 f. Remove five screws (callout 1). Figure 5-969 Remove five screws 1 Removal and replacement: DC controller 1009

g. Disconnect one connector (callout 1), and then move the guide out of the way (callout 2). Figure 5-970 Disconnect one connector 2 1 h. Slightly lift up the formatter case (callout 1), rotate the top of the case away from the chassis (callout 2), and then lift up the case to release it (callout 3). CAUTION: Pass the FFC through the opening (callout 4) in the formatter while removing the case. Figure 5-971 Release the formatter case 2 3 1 4 i. Remove the formatter case. NOTE: WiFi models only: If the formatter case is being replaced, make sure that the WiFi PCA and bracket (callout 1) are transferred to the replacement case. 1010 Chapter 5 Removal and replacement

Figure 5-972 Remove the formatter case 1 20. Remove the DC controller (DCC). CAUTION: Never remove and then install a replacement DCC and formatter at the same time. Doing so will cause the printer to become unstable. If a replacement DCC and formatter must be installed, first remove and install the formatter. Test operate the printer, and then if necessary install a replacement DCC. a. Disconnect all of the connectors and flat-flexible cables (FFCs) on the DC controller (DCC). NOTE: The number and type of connectors and FFCs depends on the printer model. A SFP printer is shown in this section. However, the procedure is correct for all printer models. Figure 5-973 Disconnect all connectors and FFCs Removal and replacement: DC controller 1011

b. Remove three screws (callout 1), and then remove the DCC (callout 2). Figure 5-974 Remove the DCC 2 1 21. Unpack the replacement assembly. a. Dispose of the defective part. NOTE: HP recommends responsible disposal of the defective part. <http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> b. Unpack the replacement part from the packaging. CAUTION: Some parts are sensitive to electrostatic discharge (ESD). Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. If the replacement part is a roller or pad, avoid touching the spongy part of it. Skin oils on the roller or pad can cause paper pickup and/or print-quality problems. HP recommends using disposable gloves when handling rollers or pads or washing your hands before touching rollers or pads. IMPORTANT: Make sure that all of the shipping materials (for example shipping tape) are removed from the replacement part prior to installation. c. To install an assembly, reverse the removal steps. NOTE: When applicable, special installation instructions are provided for an assembly at the end of the removal procedure. Always completely read the removal instructions and follow all special installation instructions. Removal and replacement: High-voltage power supply Learn how to remove and replace the high-voltage power supply (HVPS). 1012 Chapter 5 Removal and

replacement

To view a short video of this procedure, click the link below. View a video of how to remove and replace the HVPS. Mean time to repair: 12 minutes Service level: Easy

## Before performing service

### Turn the printer power off

■ Disconnect the power cable. **WARNING!** To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. Use the table below to identify the correct part number for your printer. To order the part, go to [www.hp.com/buy/parts](http://www.hp.com/buy/parts). Table 5-41 Part information

### Part number Part description

RM2-7945-000CN High-voltage power supply (HVPS)

### Required tools

● Small flat-blade screwdriver ● #2 Phillips screwdriver with a magnetic tip and a 152 mm (6 in) shaft length

## After performing service

### Turn the printer power on

■ Connect the power cable. ■ Use the power switch to turn the power on.

### Post service test

Make sure that the printer initializes to a Ready state. 1. Remove the tray. Removal and replacement: High-voltage power supply 1013

a. Pull the tray straight out of the printer until it stops. Figure 5-975 Pull the tray out until it stops b. Lift the front of the tray, and then pull it out of the printer to remove it. NOTE: If the tray has Fixed Tray Guides installed and are still required, move the hardware to new tray. If necessary, order new kit: 3GY19-67901 Kit-Fixed Tray Guide. Only one kit required per engine. Figure 5-976 Release and remove the tray 2. Remove the cartridge door assembly. NOTE: If the toner cartridge was not removed prior to servicing the printer, remove it now. 1014 Chapter 5 Removal and replacement

a. Release the support shaft (callout 1), and then slide it toward the center of the door (callout 2). TIP: It might be easier to release the support shaft by using a small flat-blade screwdriver. Figure 5-977 Release the support shaft 1 2 b. Remove the support shaft. Figure 5-978 Remove the support shaft Removal and replacement: High-voltage power supply 1015

c. Release one tab (callout 1), and then slide the guide out of the printer (callout 2). CAUTION: Be careful, the tab (callout 1) can be easily broken. Figure 5-979 Remove the guide 2 1 d. Use a small flat-blade screwdriver to gently pry the white-plastic door arm (callout 1) off of the gray-plastic pressure-release link (callout 2) to release the door arm. Figure 5-980 Release the pressure-release link 1 2 1016 Chapter 5 Removal and replacement

e. With the door in the partially closed position, slightly raise the left end (callout 1; raise it over the sheet-metal hinge boss), and then slide the door to the left (callout 2) to release the right hinge (callout 3). CAUTION: The door is still attached to the printer by the left-side retainer arm. Do not attempt to completely remove the door. Reinstallation tip: When reinstalling the door, there are two right-side hinge pins (not shown in callout 3). Keep a finger pressed on the left-side retainer arm so that it does not disengage from the door. Figure 5-981 Release the door 3 1 2 f. Rotate the door away from the printer until the keyed pin on the left-side retainer arm can disengage from the door. Figure 5-982 Release the left-side retainer arm Removal and replacement: High-voltage power supply 1017

g. Remove the door. Figure 5-983 Remove the door 3. Remove the rear door assembly. a. At the rear of the printer, remove two screws (callout 1). Figure 5-984 Remove two screws 1 1018 Chapter 5 Removal and replacement

b. Rotate the bottom edge of the assembly away from the printer to release it. Figure 5-985 Rotate the bottom of the cover c. Pull down on the assembly to remove it. Figure 5-986 Remove the cover 4. Remove the left cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. NOTE: An MFP printer is shown in this section. However, the procedure is correct for all printer models. Removal and replacement: High-voltage power supply 1019

a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-987 Locate of the tabs and bosses b. Remove one screw (callout 1). Figure 5-988 Remove one screw 1 1020 Chapter 5 Removal and replacement

c. At the front of the printer, release one boss (callout 1), and then rotate the front edge of the cover away from the printer to release three tabs (callout 2). Figure 5-989 Release one boss and three tabs 2 1 d. At the bottom edge of the cover, release two tabs. Figure 5-990 Release two tabs e. Continue to rotate the front edge of the cover away from the printer (callout 1), and then slide the cover toward the rear of the printer (callout 2) to remove it. NOTE: If the cover is difficult to rotate or slide as shown below, make sure that the bottom tabs (released in the previous step) are completely disengaged from the chassis. Removal and replacement: High-voltage power supply 1021

Figure 5-991 Remove the cover 1 2 5. Remove the HVPS. a. Disconnect one FFC (callout 1), five connectors (callout 2), two wire-harness lugs (callout 3), and then release the wire harness from the retainers (callout 4). NOTE: An MFP printer is shown in this section. However, the

procedure is correct for all printer models. The microswitch (callout 3) for the M506/E50045/M507/E50145 only has two wire-harness terminals. Figure 5-992 Disconnect the connectors 1 2 3 4 1022 Chapter 5 Removal and replacement

b. Remove two screws (callout 1). Figure 5-993 Remove two screws 1 c. Release one tab. Figure 5-994 Release one tab d. Remove the HVPS. Reinstallation tip: When the HVPS is reinstalled, make sure that the springs on the holder are positioned in the slots on the PCA, and that the PCA is behind the tabs on the holder. Removal and replacement: High-voltage power supply 1023

Figure 5-995 Remove the HVPS e. One spring behind the HVPS is not captive. Do not lose the spring when removing the HVPS. Figure 5-996 Check the spring 6. Unpack the replacement assembly. a. Dispose of the defective part. NOTE: HP recommends responsible disposal of the defective part. <http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> 1024 Chapter 5 Removal and replacement

b. Unpack the replacement part from the packaging. CAUTION: Some parts are sensitive to electrostatic discharge (ESD). Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. If the replacement part is a roller or pad, avoid touching the spongy part of it. Skin oils on the roller or pad can cause paper pickup and/or print-quality problems. HP recommends using disposable gloves when handling rollers or pads or washing your hands before touching rollers or pads. IMPORTANT: Make sure that all of the shipping materials (for example shipping tape) are removed from the replacement part prior to installation. c. To install an assembly, reverse the removal steps. NOTE: When applicable, special installation instructions are provided for an assembly at the end of the removal procedure. Always completely read the removal instructions and follow all special installation instructions. Removal and replacement: Low-voltage power supply Learn how to remove and replace the low-voltage power supply (LVPS). To view a short video of this procedure, click the link below. M501 View a video of how to remove and replace the LVPS (M051).

## M506/E50045

View a video of how to remove and replace the LVPS (M506/E50045).

## M507/E50145

View a video of how to remove and replace the LVPS (M507/E50145).

## M527/E52545/M528/E52645

View a video of how to remove and replace the LVPS (M527/E52545/M528/E52645). Mean time to repair: 12 minutes Service level: Easy

### Before performing service

#### Turn the printer power off

■ Disconnect the power cable. WARNING! To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. Removal and replacement: Low-voltage power supply 1025

Use the table below to identify the correct part number for your printer. To order the part, go to [www.hp.com/buy/parts](http://www.hp.com/buy/parts). Table 5-42 Part information

#### Part number Part description

RM2-7951-000CN Low-voltage power supply (M527/E52545/M528/E52645; 110V-127V) RM2-7952-000CN Low-voltage power supply (M527/E52545/M528/E52645; 220V-240V) RM2-7941-000CN Low-voltage power supply (M501/M506/E50045/M507/E50145; 110V-127V) RM2-7942-000CN Low-voltage power supply (M501/M506/E50045/M507/E50145; 220V-240V)

#### Required tools

- Small flat-blade screwdriver
- #2 Phillips screwdriver with a magnetic tip and a 152 mm (6 in) shaft length

### After performing service

#### Turn the printer power on

- Connect the power cable.
- Use the power switch to turn the power on.

#### Post service test

Make sure that the printer initializes to a Ready state. 1. Remove the control panel (M506x/E50045x/M507x/E50145x). 1026 Chapter 5 Removal and replacement

a. Tilt the control panel to the fully raised position. Figure 5-997 Tilt the control panel up b. Use the small flat-blade screwdriver supplied in the kit to release the control-panel screws cover, and then remove the cover. TIP: The screwdriver has a reversible shaft. Simply pull the shaft out



of the handle, and then reinstall it with the desired end facing out. Figure 5-998 Remove the cover Removal and replacement: Low-voltage power supply 1027

c. Use the small Phillips screwdriver supplied in the kit to remove two screws (callout 1). **IMPORTANT:** Store the screws in a secure place where they cannot fall down into the printer. Figure 5-999 Remove two screws 1 d. Turn the control panel over (callout 1), to gain access to the bottom side. Figure 5-1000 Turn the control panel over 1 1028 Chapter 5 Removal and replacement

e. M506/E50045 only: Disconnect one HDMI cable (callout 1), one connector (callout 2), and then remove the control panel. Figure 5-1001 Remove the control panel (M506/E50045) 1 2 f. M507/E50154 only: If the formatter cover has not been removed, remove it now. Disconnect one flat-flexible cable (FFC). **CAUTION:** ESD sensitive part. **NOTE:** This connector is a zero-insertion force (ZIF) connector. Open the locking clip. The clip is easily dislodged and lost. Figure 5-1002 Disconnect one FFC (M507/E50145) Removal and replacement: Low-voltage power supply 1029

g. M507/E50145 only: Pass the FFC through the opening in the cover, and then remove the control panel. **CAUTION:** ESD sensitive part. Figure 5-1003 Remove the control panel (M507/E50145) 2. Remove the tray. a. Pull the tray straight out of the printer until it stops. Figure 5-1004 Pull the tray out until it stops 1030 Chapter 5 Removal and replacement

b. Lift the front of the tray, and then pull it out of the printer to remove it. **NOTE:** If the tray has Fixed Tray Guides installed and are still required, move the hardware to new tray. If necessary, order new kit: 3GY19-67901 Kit-Fixed Tray Guide. Only one kit required per engine. Figure 5-1005 Release and remove the tray 3. Remove the formatter cover. ■ Do one of the following: ● M501: Remove one screw (callout 1), and then slide the cover toward the rear of the printer to remove it (callout 2). Figure 5-1006 Remove the formatter cover (M501) 2 1 ● M506/E50045/M507/E50145/M527/E52545/M528/E52645: Slide the cover toward the rear of the printer to remove it. Removal and replacement: Low-voltage power supply 1031

Figure 5-1007 Remove the formatter cover (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

4. Remove the hard-disk drive (HDD models). **CAUTION:** ESD sensitive part. a. Release the locking connector (callout 1), and then pinch the retainer (callout 2) to release it. Figure 5-1008 Release the HDD 1 2 1032 Chapter 5 Removal and replacement

b. Rotate the connector end of the HDD out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-1009 Remove the HDD 1 2 5. Remove the fax PCA (M527/E52545/M528/E52645 fax models). **CAUTION:** ESD-sensitive part. a. Disconnect three connectors (callout 1). Figure 5-1010 Disconnect three connectors 1 Removal and replacement: Low-voltage power supply 1033

b. Pinch the retainer to release it. Figure 5-1011 Release the fax PCA c. Rotate the connector end of the fax PCA out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-1012 Remove the fax PCA 1 2 1034 Chapter 5 Removal and replacement

6. Remove the formatter. **CAUTION:** Never remove and then install a replacement DCC and formatter at the same time. Doing so will cause the printer to become unstable. If a replacement DCC and formatter must be installed, first remove and install the formatter. Test operate the printer, and then if necessary install a replacement DCC. **IMPORTANT:** Take note of all empty connectors on the PCA before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessory items. a. M501: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. Figure 5-1013 Remove the formatter (M501) 1 b.

M506/E50045/M507/E50145/M527/E52545/M528/E52645: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. **IMPORTANT:** Take note of all empty connectors on the formatter before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessories items. Removal and replacement: Low-voltage power supply 1035

Figure 5-1014 Remove the formatter (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

1 1 7. Remove the stapler door or stapler blank cover (M527/E52545/M528/E52645). a. Do one of the following: ● Stapler models: Open the stapler door. ● Non-stapler models: Use a small flat-blade screw driver to release two tabs (callout 1), and then rotate the cover away from the printer (callout 2). Figure 5-1015 Open the door or release the cover 2 1 b. Do one of the following: ● Stapler models: Use a small flat-blade screwdriver to release the upper hinge. ● Non-stapler models: Slide the cover toward the front of the printer to remove it. 1036 Chapter 5 Removal and replacement

Figure 5-1016 Release the upper hinge or remove the cover c. Stapler models: Use a small flat-blade screwdriver to release the lower hinge. Figure 5-1017 Release the lower hinge Removal and replacement: Low-voltage power supply 1037

d. Stapler models: Remove the door. Figure 5-1018 Remove the door 8. Remove the stapler stationary (inner) cover (M527/E52545/M528/E52645). **NOTE:** Stapler models only. a. Before proceeding, take note of the locations of the upper tab (callout 1) and the lower tab (callout 2) on the stapler stationary (inner) cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs. Figure 5-1019 Locate the tabs 1 2 1038 Chapter 5 Removal and replacement

b. Use a long-shaft small flat-blade screwdriver to release the lower tab. Figure 5-1020 Release one tab c. Use a long-shaft small flat-blade screwdriver to release the upper tab. Figure 5-1021 Release one tab Removal and replacement: Low-voltage power supply 1039

d. With the tabs disengaged the front of the cover is released. Rotate the front of the cover away from the printer (callout 1) to release the rear of the cover (callout 2), and then remove it. Figure 5-1022 Remove the cover 1 2 9. Remove the top-left cover (M527/E52545M528/E52645).  
a. Open the toner-cartridge door, release two tabs (callout 1), and then rotate the rear edge of the cover away from the printer. Figure 5-1023 Release two tabs 1 1040 Chapter 5 Removal and replacement

b. Continue to rotate the cover away from the printer to release two more tabs at the middle of the cover. Figure 5-1024 Release two tabs c. Slide the cover toward the rear of the printer to release two bosses at the front of the cover. Figure 5-1025 Slide the cover Removal and replacement: Low-voltage power supply 1041

d. Remove the cover. Figure 5-1026 Remove the cover 10. Remove the top-rear cover (M527/E52545/M528/E52645). a. Remove one screw (callout 1), and then open the rear door. Figure 5-1027 Remove one screw 1 1042 Chapter 5 Removal and replacement

b. Rotate the exposed edge of the cover away from the printer (as shown). Figure 5-1028 Rotate the cover c. Continue to rotate the cover away from the printer to release one boss. Figure 5-1029 Release one boss Removal and replacement: Low-voltage power supply 1043

d. Remove the cover. Figure 5-1030 Remove the cover 11. Remove the rear door assembly. a. At the rear of the printer, remove two screws (callout 1). Figure 5-1031 Remove two screws 1 1044 Chapter 5 Removal and replacement

b. Rotate the bottom edge of the assembly away from the printer to release it. Figure 5-1032 Rotate the bottom of the cover c. Pull down on the assembly to remove it. Figure 5-1033 Remove the cover 12. Remove the cartridge door assembly. NOTE: If the toner cartridge was not removed prior to servicing the printer, remove it now. a. Release the support shaft (callout 1), and then slide it toward the center of the door (callout 2). TIP: It might be easier to release the support shaft by using a small flat-blade screwdriver. Removal and replacement: Low-voltage power supply 1045

Figure 5-1034 Release the support shaft 1 2 b. Remove the support shaft. Figure 5-1035 Remove the support shaft c. Release one tab (callout 1), and then slide the guide out of the printer (callout 2). CAUTION: Be careful, the tab (callout 1) can be easily broken. 1046 Chapter 5 Removal and replacement

Figure 5-1036 Remove the guide 2 1 d. Use a small flat-blade screwdriver to gently pry the white-plastic door arm (callout 1) off of the gray-plastic pressure-release link (callout 2) to release the door arm. Figure 5-1037 Release the pressure-release link 1 2 e. With the door in the partially closed position, slightly raise the left end (callout 1; raise it over the sheet-metal hinge boss), and then slide the door to the left (callout 2) to release the right hinge (callout 3). CAUTION: The door is still attached to the printer by the left-side retainer arm. Do not attempt to completely remove the door. Reinstallation tip: When reinstalling the door, there are two right-side hinge pins (not shown in callout 3). Keep a finger pressed on the left-side retainer arm so that it does not disengage from the door. Removal and replacement: Low-voltage power supply 1047

Figure 5-1038 Release the door 3 1 2 f. Rotate the door away from the printer until the keyed pin on the left-side retainer arm can disengage from the door. Figure 5-1039 Release the left-side retainer arm 1048 Chapter 5 Removal and replacement

g. Remove the door. Figure 5-1040 Remove the door 13. Remove the right cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-1041 Locate the tabs and bosses Removal and replacement: Low-voltage power supply 1049

b. Remove two screws (callout 1). Figure 5-1042 Remove two screws 1 c. At the rear of the printer, release one boss. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-1043 Release one boss d. At the bottom edge of the cover, release two bosses. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. 1050 Chapter 5 Removal and replacement

Figure 5-1044 Release two bosses e. Near the formatter, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Hold the bottom of the cover away from the printer, and then use a small flat-blade screw driver to easily release this tab. Figure 5-1045 Release one tab Removal and replacement: Low-voltage power supply 1051

f. At the top rear edge of the cover, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Use a small flat-blade screw driver to easily release this tab. Figure 5-1046 Release one tab g. Rotate the rear of the cover away from the printer (callout 1), slide the cover toward the front of the printer (callout 2) to release two bosses (callout 3), and then remove the cover. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-1047 Remove the cover 2 1 3 1052 Chapter 5 Removal and replacement

14. Remove the left cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. NOTE: An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-1048 Locate of the tabs and bosses b. Remove one screw (callout 1). Figure 5-1049 Remove one screw 1 Removal and replacement: Low-voltage power supply 1053

c. At the front of the printer, release one boss (callout 1), and then rotate the front edge of the cover away from the printer to release three tabs (callout 2). Figure 5-1050 Release one boss and three tabs 2 1 d. At the bottom edge of the cover, release two tabs. Figure 5-1051 Release two tabs e. Continue to rotate the front edge of the cover away from the printer (callout 1), and then slide the cover toward the rear of the printer (callout 2) to remove it. NOTE: If the cover is difficult to rotate or slide as shown below, make sure that the bottom tabs (released in the previous step) are completely disengaged from the chassis. 1054 Chapter 5 Removal and replacement

Figure 5-1052 Remove the cover 1 2 15. Remove the integrated-scanner assembly (ISA) (M527/E52545/M528/E52645). NOTE: The ISA includes the document feeder, sub-scanner assembly (SSA), and the scanner control board (SCB). a. Remove one screw (callout 1). IMPORTANT: This screw fastens the ISA ground plate to the printer chassis. Do not lose this screw. Figure 5-1053 Remove one screw 1 Removal and replacement: Low-voltage power supply 1055

b. Disconnect three connectors (callout 1), pass the wire harnesses and cables through the opening in the cover (callout 2), and then release them from the retainers (callout 3). NOTE: One cable (callout 4) does not need to pass through the opening in the cover. TIP: It might be easier if the WiFi PCA and holder (not all models; callout 5) is detached from the printer, and then moved out of the way. Figure 5-1054 Disconnect cables and connectors 4 1 2 3 5 c. Disconnect three connectors (callout 1), and then remove six screws (callout 2). Figure 5-1055 Disconnect connectors and remove screws 1 2 1056 Chapter 5 Removal and replacement

d. Slightly slide the ISA towards the back of the printer to release it. Figure 5-1056 Slide the ISA towards the back of the printer e. Remove the ISA. Figure 5-1057 Remove the ISA 16. Remove the top cover (M527/E52545/M528/E52645). Removal and replacement: Low-voltage power supply 1057

a. At the front right of the printer, remove one screw (callout 1), and then remove the sheet-metal plate (callout 2). Figure 5-1058 Remove one screw and the plate 1 2 b. At the top rear of the printer, remove three screws (callout 1). Figure 5-1059 Remove three screws 1 c. At the top front of the printer, remove three screws (callout 1). NOTE: The screw (callout 2) on the lower-right portion of the cover is a different color than the other two screws, and is surrounded by sheet-metal. It might appear that this screw does not need to be removed. Remove this screw. 1058 Chapter 5 Removal and replacement

Figure 5-1060 Remove three screws 1 2 d. At the front left of the printer, release one boss. Figure 5-1061 Release one boss e. Rotate the front of the cover up and away from the printer. CAUTION: As the cover is rotated up, carefully pass the cables and wire harnesses the hole in the cover (callout 1). Removal and replacement: Low-voltage power supply 1059

Figure 5-1062 Rotate the front of the cover up 1 f. Remove the cover. Figure 5-1063 Remove the cover 17. Remove the top cover (M501/M506/E50045/M507/E50145). The M506x/E50045x is shown in this procedure. However, the steps are correct for all M506/E50045/M507/E50145 models and the M501 printer. 1060 Chapter 5 Removal and replacement

a. Remove two screws (callout 1). Figure 5-1064 Remove two screws 1 b. M506/E50045/ only: Release one cable retainer from the chassis. TIP: Depress the tab on the retainer to release it. Figure 5-1065 Release one cable retainer Removal and replacement: Low-voltage power supply 1061

c. M507/E50145 only: Disconnect two connectors. Figure 5-1066 Disconnect two connectors d. Release one boss. Figure 5-1067 Release one boss e. M506/E50045/ only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M506x/E50045x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. 1062 Chapter 5 Removal and replacement

Figure 5-1068 Remove the cover (M506/E50045) 1 f. M507/E50145only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M507x/E50145x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). IMPORTANT: If the top cover is being replaced, make sure to transfer the WiFi PCA from the discarded cover to the replacement cover. NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. Figure 5-1069 Remove the cover (M507/E50145) 1 Removal and replacement: Low-voltage power supply 1063

18. Remove the paper delivery tray (output bin). NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. a. Release two tabs (callout 1), and then slide the cover toward the front of the printer (callout 2). Figure 5-1070 Release two tabs 1 2 1 b. Remove the bin. Figure 5-1071 Remove the bin 19. Remove the formatter case. 1064 Chapter 5 Removal and replacement

a. M506/E50045/M507/E50145 only: Remove two screws (callout 1), and then remove the sheet metal plate (callout 2). NOTE: M507/E50145 only: The M507/E50145 sheet-metal plate includes a separate black plastic cable guide (not shown). Do not lose the guide when removing the plate. Figure 5-1072 Remove the sheet-metal plate 2 1 b. Carefully pass all cables and wire harnesses through the opening in the formatter case. NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. Figure 5-1073 Pass all cables and wire harnesses through the opening Removal and replacement: Low-voltage power supply 1065

c. Remove one screw (callout 1), and then push down on one tab (callout 2) to release it. Figure 5-1074 Remove one screw 1 2 d. Remove the guide. Figure 5-1075 Remove the guide 1066 Chapter 5 Removal and replacement

e. Remove four screws (callout 1). Figure 5-1076 Remove four screws 1 f. Remove five screws (callout 1). Figure 5-1077 Remove five screws 1 Removal and replacement: Low-voltage power supply 1067

g. Disconnect one connector (callout 1), and then move the guide out of the way (callout 2). Figure 5-1078 Disconnect one connector 2 1 h. Slightly lift up the formatter case (callout 1), rotate the top of the case away from the chassis (callout 2), and then lift up the case to release it (callout 3). CAUTION: Pass the FFC through the opening (callout 4) in the formatter while removing the case. Figure 5-1079 Release the formatter case 2 3 1 4 i. Remove the formatter case. NOTE: WiFi models only: If the formatter case is being replaced, make sure that the WiFi PCA and bracket (callout 1) are transferred to the replacement case. 1068 Chapter 5 Removal and replacement

Figure 5-1080 Remove the formatter case 1 20. Remove the LVPS. a. Remove one screw (callout 1), and then move the power-switch assembly (callout 2) out of the way. Figure 5-1081 Move the power-switch assembly 1 2 Removal and replacement: Low-voltage power supply 1069

b. Disconnect one connector (callout 1), and then release the wire harness from the guide (callout 2). Figure 5-1082 Disconnect one connector 1 2 c. Disconnect two connectors (callout 1). TIP: Squeeze the locking lever on the lower connector to release it. Figure 5-1083 Disconnect two connectors 1 1070 Chapter 5 Removal and replacement

d. Remove two screws (callout 1). Figure 5-1084 Remove two screws 1 e. Release one tab (callout 1), and then rotate the top of the LVPS out and away from the printer (callout 2). Figure 5-1085 Release one tab 1 2 f. Slide the LVPS up and away from the printer to release it.

Reinstallation tip: Push down on the LVPS to make sure that it is fully seated on the holder. Removal and replacement: Low-voltage power supply 1071

Figure 5-1086 Release the LVPS g. Remove the LVPS. Figure 5-1087 Remove the LVPS 21. Unpack the replacement assembly. a. Dispose of the defective part. NOTE: HP recommends responsible disposal of the defective part.

<http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> 1072 Chapter 5 Removal and replacement

b. Unpack the replacement part from the packaging. CAUTION: Some parts are sensitive to electrostatic discharge (ESD). Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. If the replacement part is a roller or pad, avoid touching the spongy part of it. Skin oils on the roller or pad can cause paper pickup and/or print-quality problems. HP recommends using disposable gloves when handling rollers or pads or washing your hands before touching rollers or pads. IMPORTANT: Make sure that all of the shipping materials (for example shipping tape) are removed from the replacement part prior to installation. c. To install an assembly, reverse the removal steps. NOTE: When applicable, special installation instructions are provided for an assembly at the end of the removal procedure. Always completely read the removal instructions and follow all special installation instructions. Removal and replacement: Main Motor M1 Learn how to remove and replace the Main Motor M1. IMPORTANT: It is recommended to bring the printer back to the shop to perform the following removal and replacement repairs. Mean time to repair: 60 minutes Service level: Hard

## Before performing service

### Turn the printer power off

■ Disconnect the power cable. WARNING! To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. Use the table below to identify the correct part number for your printer. To order the part, go to [www.hp.com/buy/parts](http://www.hp.com/buy/parts). Table 5-43 Part information

### Part number Part description

RM2-8684-000CN Main Motor M1

### Required tools

● Small flat-blade screwdriver ● #2 Phillips screwdriver with a magnetic tip and a 152 mm (6 in) shaft length Removal and replacement: Main Motor M1 1073

## After performing service

### Turn the printer power on

■ Connect the power cable. ■ Use the power switch to turn the power on.

### Post service test

Make sure that the printer initializes to a Ready state. 1. Remove the control panel (M506x/E50045x/M507x/E50145x). a. Tilt the control panel to the fully raised position. Figure 5-1088 Tilt the control panel up b. Use the small flat-blade screwdriver supplied in the kit to release the control-panel screws cover, and then remove the cover. TIP: The screwdriver has a reversible shaft. Simply pull the shaft out of the handle, and then reinstall it with the desired end facing out. 1074 Chapter 5 Removal and replacement

Figure 5-1089 Remove the cover c. Use the small Phillips screwdriver supplied in the kit to remove two screws (callout 1). IMPORTANT: Store the screws in a secure place where they cannot fall down into the printer. Figure 5-1090 Remove two screws 1 Removal and replacement: Main Motor M1 1075

d. Turn the control panel over (callout 1), to gain access to the bottom side. Figure 5-1091 Turn the control panel over 1 e. M506/E50045 only: Disconnect one HDMI cable (callout 1), one connector (callout 2), and then remove the control panel. Figure 5-1092 Remove the control panel (M506/E50045) 1 2 f. M507/E50145 only: If the formatter cover has not been removed, remove it now. Disconnect one flat-flexible cable (FFC). CAUTION: ESD sensitive part. NOTE: This connector is a zero-insertion force (ZIF) connector. Open the locking clip. The clip is easily dislodged and lost. 1076 Chapter 5 Removal and replacement

Figure 5-1093 Disconnect one FFC (M507/E50145) g. M507/E50145 only: Pass the FFC through the opening in the cover, and then remove the control panel. CAUTION: ESD sensitive part. Figure 5-1094 Remove the control panel (M507/E50145) 2. Remove the tray. Removal and replacement: Main Motor M1 1077

a. Pull the tray straight out of the printer until it stops. Figure 5-1095 Pull the tray out until it stops b. Lift the front of the tray, and then pull it out of the printer to remove it. NOTE: If the tray has Fixed Tray Guides installed and are still required, move the hardware to new tray. If necessary, order new kit: 3GY19-67901 Kit-Fixed Tray Guide. Only one kit required per engine. Figure 5-1096 Release and remove the tray 3. Remove the formatter cover. ■ Do one of the following: ● M501: Remove one screw (callout 1), and then slide the cover toward the rear of the printer to remove it (callout 2). 1078 Chapter 5 Removal and replacement

Figure 5-1097 Remove the formatter cover (M501) 2 1 ● M506/E50045/M507/E50145/M527/E52545/M528/E52645: Slide the cover toward the rear of the printer to remove it. Figure 5-1098 Remove the formatter cover (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

4. Remove the hard-disk drive (HDD models). CAUTION: ESD sensitive part. Removal and replacement: Main Motor M1 1079

a. Release the locking connector (callout 1), and then pinch the retainer (callout 2) to release it. Figure 5-1099 Release the HDD 1 2 b. Rotate the connector end of the HDD out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-1100 Remove the HDD 1 2 5. Remove the fax PCA (M527/E52545/M528/E52645 fax models). CAUTION: ESD-sensitive part. 1080 Chapter 5 Removal and replacement

a. Disconnect three connectors (callout 1). Figure 5-1101 Disconnect three connectors 1 b. Pinch the retainer to release it. Figure 5-1102 Release the fax PCA Removal and replacement: Main Motor M1 1081

c. Rotate the connector end of the fax PCA out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-1103 Remove the fax PCA 1 2 6. Remove the formatter. CAUTION: Never remove and then install a replacement DCC and formatter at the same time. Doing so will cause the printer to become unstable. If a replacement DCC and formatter must be installed, first remove and install the formatter. Test operate the printer, and then if necessary install a replacement DCC. IMPORTANT: Take note of all empty connectors on the PCA before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessory items. 1082 Chapter 5 Removal and replacement

a. M501: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. Figure 5-1104 Remove the formatter (M501) 1 b. M506/E50045/M507/E50145/M527/E52545/M528/E52645: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. IMPORTANT: Take note of all empty connectors on the formatter before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessories items. Figure 5-1105 Remove the formatter (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

1 1 7. Remove the stapler door or stapler blank cover (M527/E52545/M528/E52645). a. Do one of the following: ● Stapler models: Open the stapler door. ● Non-stapler models: Use a small flat-blade screw driver to release two tabs (callout 1), and then rotate the cover away from the printer (callout 2). Removal and replacement: Main Motor M1 1083

Figure 5-1106 Open the door or release the cover 2 1 b. Do one of the following: ● Stapler models: Use a small flat-blade screwdriver to release the upper hinge. ● Non-stapler models: Slide the cover toward the front of the printer to remove it. Figure 5-1107 Release the upper hinge or remove the cover 1084 Chapter 5 Removal and replacement

c. Stapler models: Use a small flat-blade screwdriver to release the lower hinge. Figure 5-1108 Release the lower hinge d. Stapler models: Remove the door. Figure 5-1109 Remove the door 8. Remove the stapler stationary (inner) cover (M527/E52545/M528/E52645). NOTE: Stapler models only. a. Before proceeding, take note of the locations of the upper tab (callout 1) and the lower tab (callout 2) on the stapler stationary (inner) cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs. Removal and replacement: Main Motor M1 1085

Figure 5-1110 Locate the tabs 1 2 b. Use a long-shaft small flat-blade screwdriver to release the lower tab. Figure 5-1111 Release one tab 1086 Chapter 5 Removal and replacement

c. Use a long-shaft small flat-blade screwdriver to release the upper tab. Figure 5-1112 Release one tab d. With the tabs disengaged the front of the cover is released. Rotate the front of the cover away from the printer (callout 1) to release the rear of the cover (callout 2), and then remove it. Figure 5-1113 Remove the cover 1 2 9. Remove the top-left cover (M527/E52545/M528/E52645). Removal and replacement: Main Motor M1 1087

a. Open the toner-cartridge door, release two tabs (callout 1), and then rotate the rear edge of the cover away from the printer. Figure 5-1114 Release two tabs 1 b. Continue to rotate the cover away from the printer to release two more tabs at the middle of the cover. Figure 5-1115 Release two tabs 1088 Chapter 5 Removal and replacement

c. Slide the cover toward the rear of the printer to release two bosses at the front of the cover. Figure 5-1116 Slide the cover d. Remove the cover. Figure 5-1117 Remove the cover 10. Remove the top-rear cover (M527/E52545/M528/E52645). Removal and replacement: Main Motor M1 1089

a. Remove one screw (callout 1), and then open the rear door. Figure 5-1118 Remove one screw 1 b. Rotate the exposed edge of the cover away from the printer (as shown). Figure 5-1119 Rotate the cover 1090 Chapter 5 Removal and replacement

c. Continue to rotate the cover away from the printer to release one boss. Figure 5-1120 Release one boss d. Remove the cover. Figure 5-1121 Remove the cover 11. Remove the rear door assembly. Removal and replacement: Main Motor M1 1091

a. At the rear of the printer, remove two screws (callout 1). Figure 5-1122 Remove two screws 1 b. Rotate the bottom edge of the assembly away from the printer to release it. Figure 5-1123 Rotate the bottom of the cover 1092 Chapter 5 Removal and replacement

c. Pull down on the assembly to remove it. Figure 5-1124 Remove the cover 12. Remove the cartridge door assembly. NOTE: If the toner cartridge was not removed prior to servicing the printer, remove it now. a. Release the support shaft (callout 1), and then slide it toward the center of the door (callout 2). TIP: It might be easier to release the support shaft by using a small flat-blade screwdriver. Removal and replacement: Main Motor M1 1093

Figure 5-1125 Release the support shaft 1 2 b. Remove the support shaft. Figure 5-1126 Remove the support shaft c. Release one tab (callout 1), and then slide the guide out of the printer (callout 2). CAUTION: Be careful, the tab (callout 1) can be easily broken. 1094 Chapter

## 5 Removal and replacement

Figure 5-1127 Remove the guide 2 1 d. Use a small flat-blade screwdriver to gently pry the white-plastic door arm (callout 1) off of the gray-plastic pressure-release link (callout 2) to release the door arm. Figure 5-1128 Release the pressure-release link 1 2 e. With the door in the partially closed position, slightly raise the left end (callout 1; raise it over the sheet-metal hinge boss), and then slide the door to the left (callout 2) to release the right hinge (callout 3). CAUTION: The door is still attached to the printer by the left-side retainer arm. Do not attempt to completely remove the door. Reinstallation tip: When reinstalling the door, there are two right-side hinge pins (not shown in callout 3). Keep a finger pressed on the left-side retainer arm so that it does not disengage from the door. Removal and replacement: Main Motor M1 1095

Figure 5-1129 Release the door 3 1 2 f. Rotate the door away from the printer until the keyed pin on the left-side retainer arm can disengage from the door. Figure 5-1130 Release the left-side retainer arm 1096 Chapter 5 Removal and replacement

g. Remove the door. Figure 5-1131 Remove the door 13. Remove the right cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-1132 Locate the tabs and bosses Removal and replacement: Main Motor M1 1097

b. Remove two screws (callout 1). Figure 5-1133 Remove two screws 1 c. At the rear of the printer, release one boss. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-1134 Release one boss d. At the bottom edge of the cover, release two bosses. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. 1098 Chapter 5 Removal and replacement

Figure 5-1135 Release two bosses e. Near the formatter, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Hold the bottom of the cover away from the printer, and then use a small flat-blade screw driver to easily release this tab. Figure 5-1136 Release one tab Removal and replacement: Main Motor M1 1099

f. At the top rear edge of the cover, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Use a small flat-blade screw driver to easily release this tab. Figure 5-1137 Release one tab g. Rotate the rear of the cover away from the printer (callout 1), slide the cover toward the front of the printer (callout 2) to release two bosses (callout 3), and then remove the cover. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-1138 Remove the cover 2 1 3 1100 Chapter 5 Removal and replacement

14. Remove the left cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. NOTE: An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-1139 Locate of the tabs and bosses b. Remove one screw (callout 1). Figure 5-1140 Remove one screw 1 Removal and replacement: Main Motor M1 1101

c. At the front of the printer, release one boss (callout 1), and then rotate the front edge of the cover away from the printer to release three tabs (callout 2). Figure 5-1141 Release one boss and three tabs 2 1 d. At the bottom edge of the cover, release two tabs. Figure 5-1142 Release two tabs e. Continue to rotate the front edge of the cover away from the printer (callout 1), and then slide the cover toward the rear of the printer (callout 2) to remove it. NOTE: If the cover is difficult to rotate or slide as shown below, make sure that the bottom tabs (released in the previous step) are completely disengaged from the chassis. 1102 Chapter 5 Removal and replacement

Figure 5-1143 Remove the cover 1 2 15. Remove the integrated-scanner assembly (ISA) (M527/E52545/M528/E52645). NOTE: The ISA includes the document feeder, sub-scanner assembly (SSA), and the scanner control board (SCB). a. Remove one screw (callout 1). IMPORTANT: This screw fastens the ISA ground plate to the printer chassis. Do not lose this screw. Figure 5-1144 Remove one screw 1 Removal and replacement: Main Motor M1 1103

b. Disconnect three connectors (callout 1), pass the wire harnesses and cables through the opening in the cover (callout 2), and then release them from the retainers (callout 3). NOTE: One cable (callout 4) does not need to pass through the opening in the cover. TIP: It might be easier if the WiFi PCA and holder (not all models; callout 5) is detached from the printer, and then moved out of the way. Figure 5-1145 Disconnect cables and connectors 4 1 2 3 5 c. Disconnect three connectors (callout 1), and then remove six screws (callout 2). Figure 5-1146 Disconnect connectors and remove screws 1 2 1104 Chapter 5 Removal and replacement

d. Slightly slide the ISA towards the back of the printer to release it. Figure 5-1147 Slide the ISA towards the back of the printer e. Remove the ISA. Figure 5-1148 Remove the ISA 16. Remove the top cover (M527/E52545/M528/E52645). Removal and replacement: Main Motor M1 1105

a. At the front right of the printer, remove one screw (callout 1), and then remove the sheet-metal plate (callout 2). Figure 5-1149 Remove one screw and the plate 1 2 b. At the top rear of the printer, remove three screws (callout 1). Figure 5-1150 Remove three screws 1 c. At the top front of the printer, remove three screws (callout 1). NOTE: The screw (callout 2) on the lower-right portion of the cover is a different color than the other two screws, and is surrounded by sheet-metal. It might appear that this screw does not need to be removed. Remove this screw. 1106 Chapter 5 Removal and replacement

Figure 5-1151 Remove three screws 1 2 d. At the front left of the printer, release one boss. Figure 5-1152 Release one boss e. Rotate the front of the cover up and away from the printer. CAUTION: As the cover is rotated up, carefully pass the cables and wire harnesses the hole in the cover (callout 1). Removal and replacement: Main Motor M1 1107

Figure 5-1153 Rotate the front of the cover up 1 f. Remove the cover. Figure 5-1154 Remove the cover 17. Remove the top cover (M501/M506/E50045/M507/E50145). The M506x/E50045x is shown in this procedure. However, the steps are correct for all M506/E50045/M507/E50145 models and the M501 printer. 1108 Chapter 5 Removal and replacement

a. Remove two screws (callout 1). Figure 5-1155 Remove two screws 1 b. M506/E50045/ only: Release one cable retainer from the chassis. TIP: Depress the tab on the retainer to release it. Figure 5-1156 Release one cable retainer Removal and replacement: Main Motor M1 1109

c. M507/E50145 only: Disconnect two connectors. Figure 5-1157 Disconnect two connectors d. Release one boss. Figure 5-1158 Release one boss e. M506/E50045/ only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M506x/E50045x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. 1110 Chapter 5 Removal and replacement

Figure 5-1159 Remove the cover (M506/E50045) 1 f. M507/E50145only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M507x/E50145x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). IMPORTANT: If the top cover is being replaced, make sure to transfer the WiFi PCA from the discarded cover to the replacement cover. NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. Figure 5-1160 Remove the cover (M507/E50145) 1 Removal and replacement: Main Motor M1 1111

18. Remove the paper delivery tray (output bin). NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. a. Release two tabs (callout 1), and then slide the cover toward the front of the printer (callout 2). Figure 5-1161 Release two tabs 1 2 1 b. Remove the bin. Figure 5-1162 Remove the bin 19. Remove the formatter case. 1112 Chapter 5 Removal and replacement

a. M506/E50045/M507/E50145 only: Remove two screws (callout 1), and then remove the sheet metal plate (callout 2). NOTE: M507/E50145 only: The M507/E50145 sheet-metal plate includes a separate black plastic cable guide (not shown). Do not lose the guide when removing the plate. Figure 5-1163 Remove the sheet-metal plate 2 1 b. Carefully pass all cables and wire harnesses through the opening in the formatter case. NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. Figure 5-1164 Pass all cables and wire harnesses through the opening Removal and replacement: Main Motor M1 1113

c. Remove one screw (callout 1), and then push down on one tab (callout 2) to release it. Figure 5-1165 Remove one screw 1 2 d. Remove the guide. Figure 5-1166 Remove the guide 1114 Chapter 5 Removal and replacement

e. Remove four screws (callout 1). Figure 5-1167 Remove four screws 1 f. Remove five screws (callout 1). Figure 5-1168 Remove five screws 1 Removal and replacement: Main Motor M1 1115

g. Disconnect one connector (callout 1), and then move the guide out of the way (callout 2). Figure 5-1169 Disconnect one connector 2 1 h. Slightly lift up the formatter case (callout 1), rotate the top of the case away from the chassis (callout 2), and then lift up the case to release it (callout 3). CAUTION: Pass the FFC through the opening (callout 4) in the formatter while removing the case. Figure 5-1170 Release the formatter case 2 3 1 4 i. Remove the formatter case. NOTE: WiFi models only: If the formatter case is being replaced, make sure that the WiFi PCA and bracket (callout 1) are transferred to the replacement case. 1116 Chapter 5 Removal and replacement

Figure 5-1171 Remove the formatter case 1 20. Remove the LVPS. a. Remove one screw (callout 1), and then move the power-switch assembly (callout 2) out of the way. Figure 5-1172 Move the power-switch assembly 1 2 Removal and replacement: Main Motor M1 1117

b. Disconnect one connector (callout 1), and then release the wire harness from the guide (callout 2). Figure 5-1173 Disconnect one connector 1 2 c. Disconnect two connectors (callout 1). TIP: Squeeze the locking lever on the lower connector to release it. Figure 5-1174 Disconnect two connectors 1 1118 Chapter 5 Removal and replacement

d. Remove two screws (callout 1). Figure 5-1175 Remove two screws 1 e. Release one tab (callout 1), and then rotate the top of the LVPS out and away from the printer (callout 2). Figure 5-1176 Release one tab 1 2 f. Slide the LVPS up and away from the printer to release it. Reinstallation tip: Push down on the LVPS to make sure that it is fully seated on the holder. Removal and replacement: Main Motor M1 1119

Figure 5-1177 Release the LVPS g. Remove the LVPS. Figure 5-1178 Remove the LVPS 21. Remove the fan (FM2). 1120 Chapter 5 Removal and replacement

a. Disconnect one connector (callout 1), and then release the wire harness from the retainer (callout 2). Figure 5-1179 Disconnect one connector 1 2 b. Release two tabs (callout 1), and then rotate the top of the fan out and away from the printer (callout 2). Figure 5-1180 Release two tabs 2 1 c. Remove the fan. CAUTION: When reinstalled the fan must draw air into the printer. Make sure that the two arrows embossed on the right side of the fan body face downward and into the printer. Removal and replacement: Main Motor M1 1121

Figure 5-1181 Remove the fan 22. Remove the fuser power supply (FPS). TIP: The FPS includes the optional paper feeder connector (Jetlink accessory tray connector) . a. Disconnect three connectors (callout 1). Figure 5-1182 Disconnect three connectors 1 1122 Chapter 5 Removal and replacement

b. Remove one screw (callout 1). Figure 5-1183 Remove one screw 1 c. Disconnect the tray accessory connector (circle callout) and release the wire harness from the guide. d. Release four tabs. TIP: It might be easier to release the upper two tabs by using a small flat-blade screw driver. Removal and replacement: Main Motor M1 1123

Figure 5-1184 Release four tabs e. Rotate the left end of the assembly out and away from the printer (callout 1), slide the assembly to the left to release the right side alignment pins (callout 2), and then remove the assembly. Figure 5-1185 Remove the assembly 1 2 23. Remove internal cable and holder brackets. IMPORTANT: It is recommended to bring the printer back to the shop to perform the following removal and replacement repairs. NOTE: Lay the engine on its left side to make removal or the following parts easier. 1124 Chapter 5 Removal and replacement

a. Remove one (1) screw from the upper cable bracket. NOTE: The upper cable bracket assembly does not need to be completely removed, simply detached and then moved up out of the way and placed on the top of the printer. b. Remove the cable routing from the upper cable bracket. c. Lift up on the left side of the cable bracket (callout 1) and slide the bracket towards the front of the printer (callout 2) to remove the bracket. 1 2 Removal and replacement: Main Motor M1 1125

d. Remove the fuser power supply bracket (small lower rear bracket) by pulling in the upper and left side of the bracket and lifting it from the printer. e. Remove the cables from the lower front cable bracket. 1126 Chapter 5 Removal and replacement



f. Remove three (3) screws (callout 1) from the bracket assembly. 1 g. Gently pry up the two (2) tabs on the lower right side of the bracket the remove the bracket from the printer. Removal and replacement: Main Motor M1 1127

h. Remove two (2) screws (callout 1) from the metal M1 motor shield and remove the shield from the printer. 1 i. Remove three (2) screws (callout 1) and then remove the cable bracket located to the right of the main motor, M1. NOTE: Carefully lift off the bracket to ensure that the clutch on the back of the guide does not fall off when removing. 1 24. Remove the Main Motor M1. IMPORTANT: It is recommended to bring the printer back to the shop to perform the following removal and replacement repairs. NOTE: Lay the engine on its left side to make removal or the following parts easier. 1128 Chapter 5 Removal and replacement

■ Remove three (3) screws from the main motor M1 and then remove the motor from the printer. 1 2 Reinstallation tip: When reinstalling the main motor, ensure that the motor board fits under the tab (Callout 2). Start all three (3) screws to hold the motor in place while adjusting the motor to ensure that it seats correctly then tighten all three screws. 25. Unpack the replacement assembly. a. Dispose of the defective part. NOTE: HP recommends responsible disposal of the defective part.

<http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> b. Unpack the replacement part from the packaging. CAUTION: Some parts are sensitive to electrostatic discharge (ESD). Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. If the replacement part is a roller or pad, avoid touching the spongy part of it. Skin oils on the roller or pad can cause paper pickup and/or print-quality problems. HP recommends using disposable gloves when handling rollers or pads or washing your hands before touching rollers or pads. IMPORTANT: Make sure that all of the shipping materials (for example shipping tape) are removed from the replacement part prior to installation. c. To install an assembly, reverse the removal steps. NOTE: When applicable, special installation instructions are provided for an assembly at the end of the removal procedure. Always completely read the removal instructions and follow all special installation instructions. Removal and replacement: Main Motor M1 1129

Removal and replacement: Paper feed drive assembly Learn how to remove and replace the Paper feed drive assembly. IMPORTANT: It is recommended to bring the printer back to the shop to perform the following removal and replacement repairs. Mean time to repair: 60 minutes Service level: Hard

## Before performing service

### Turn the printer power off

■ Disconnect the power cable. WARNING! To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. Use the table below to identify the correct part number for your printer. To order the part, go to [www.hp.com/buy/parts](http://www.hp.com/buy/parts). Table 5-44 Part information

### Part number Part description

RM2-5715-000CN Paper feed drive assembly

### Required tools

● Small flat-blade screwdriver ● #2 Phillips screwdriver with a magnetic tip and a 152 mm (6 in) shaft length

## After performing service

### Turn the printer power on

■ Connect the power cable. ■ Use the power switch to turn the power on.

### Post service test

Make sure that the printer initializes to a Ready state. 1. Remove the control panel (M506x/E50045x/M507x/E50145x). 1130 Chapter 5 Removal and replacement

a. Tilt the control panel to the fully raised position. Figure 5-1186 Tilt the control panel up b. Use the small flat-blade screwdriver supplied in the kit to release the control-panel screws cover, and then remove the cover. TIP: The screwdriver has a reversible shaft. Simply pull the shaft out of the handle, and then reinstall it with the desired end facing out. Figure 5-1187 Remove the cover Removal and replacement: Paper feed drive assembly 1131

c. Use the small Phillips screwdriver supplied in the kit to remove two screws (callout 1). IMPORTANT: Store the screws in a secure place where they cannot fall down into the printer. Figure 5-1188 Remove two screws 1 d. Turn the control panel over (callout 1), to gain access to the bottom side. Figure 5-1189 Turn the control panel over 1 1132 Chapter 5 Removal and replacement

e. M506/E50045 only: Disconnect one HDMI cable (callout 1), one connector (callout 2), and then remove the control panel. Figure 5-1190 Remove the control panel (M506/E50045) 1 2 f. M507/E50154 only: If the formatter cover has not been removed, remove it now. Disconnect one flat-flexible cable (FFC). CAUTION: ESD sensitive part. NOTE: This connector is a zero-insertion force (ZIF) connector. Open the locking clip. The clip is easily dislodged and lost. Figure 5-1191 Disconnect one FFC (M507/E50145) Removal and replacement: Paper feed drive assembly 1133

g. M507/E50145 only: Pass the FFC through the opening in the cover, and then remove the control panel. CAUTION: ESD sensitive part. Figure 5-1192 Remove the control panel (M507/E50145) 2. Remove the tray. a. Pull the tray straight out of the printer until it stops. Figure

5-1193 Pull the tray out until it stops 1134 Chapter 5 Removal and replacement

b. Lift the front of the tray, and then pull it out of the printer to remove it. NOTE: If the tray has Fixed Tray Guides installed and are still required, move the hardware to new tray. If necessary, order new kit: 3GY19-67901 Kit-Fixed Tray Guide. Only one kit required per engine. Figure 5-1194 Release and remove the tray 3. Remove the formatter cover. ■ Do one of the following: ● M501: Remove one screw (callout 1), and then slide the cover toward the rear of the printer to remove it (callout 2). Figure 5-1195 Remove the formatter cover (M501) 2 1 ● M506/E50045/M507/E50145/M527/E52545/M528/E52645: Slide the cover toward the rear of the printer to remove it. Removal and replacement: Paper feed drive assembly 1135

Figure 5-1196 Remove the formatter cover (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

4. Remove the hard-disk drive (HDD models). CAUTION: ESD sensitive part. a. Release the locking connector (callout 1), and then pinch the retainer (callout 2) to release it. Figure 5-1197 Release the HDD 1 2 1136 Chapter 5 Removal and replacement

b. Rotate the connector end of the HDD out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-1198 Remove the HDD 1 2 5. Remove the fax PCA (M527/E52545/M528/E52645 fax models). CAUTION: ESD-sensitive part. a. Disconnect three connectors (callout 1). Figure 5-1199 Disconnect three connectors 1 Removal and replacement: Paper feed drive assembly 1137

b. Pinch the retainer to release it. Figure 5-1200 Release the fax PCA c. Rotate the connector end of the fax PCA out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-1201 Remove the fax PCA 1 2 1138 Chapter 5 Removal and replacement

6. Remove the formatter. CAUTION: Never remove and then install a replacement DCC and formatter at the same time. Doing so will cause the printer to become unstable. If a replacement DCC and formatter must be installed, first remove and install the formatter. Test operate the printer, and then if necessary install a replacement DCC. IMPORTANT: Take note of all empty connectors on the PCA before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessory items. a. M501: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. Figure 5-1202 Remove the formatter (M501) 1 b.

M506/E50045/M507/E50145/M527/E52545/M528/E52645: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. IMPORTANT: Take note of all empty connectors on the formatter before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessories items. Removal and replacement: Paper feed drive assembly 1139

Figure 5-1203 Remove the formatter (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

1 1 7. Remove the stapler door or stapler blank cover (M527/E52545/M528/E52645). a. Do one of the following: ● Stapler models: Open the stapler door. ● Non-stapler models: Use a small flat-blade screw driver to release two tabs (callout 1), and then rotate the cover away from the printer (callout 2). Figure 5-1204 Open the door or release the cover 2 1 b. Do one of the following: ● Stapler models: Use a small flat-blade screwdriver to release the upper hinge. ● Non-stapler models: Slide the cover toward the front of the printer to remove it. 1140 Chapter 5 Removal and replacement

Figure 5-1205 Release the upper hinge or remove the cover c. Stapler models: Use a small flat-blade screwdriver to release the lower hinge. Figure 5-1206 Release the lower hinge Removal and replacement: Paper feed drive assembly 1141

d. Stapler models: Remove the door. Figure 5-1207 Remove the door 8. Remove the stapler stationary (inner) cover (M527/E52545/M528/E52645). NOTE: Stapler models only. a. Before proceeding, take note of the locations of the upper tab (callout 1) and the lower tab (callout 2) on the stapler stationary (inner) cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs. Figure 5-1208 Locate the tabs 1 2 1142 Chapter 5 Removal and replacement

b. Use a long-shaft small flat-blade screwdriver to release the lower tab. Figure 5-1209 Release one tab c. Use a long-shaft small flat-blade screwdriver to release the upper tab. Figure 5-1210 Release one tab Removal and replacement: Paper feed drive assembly 1143

d. With the tabs disengaged the front of the cover is released. Rotate the front of the cover away from the printer (callout 1) to release the rear of the cover (callout 2), and then remove it. Figure 5-1211 Remove the cover 1 2 9. Remove the top-left cover (M527/E52545/M528/E52645).

a. Open the toner-cartridge door, release two tabs (callout 1), and then rotate the rear edge of the cover away from the printer. Figure 5-1212 Release two tabs 1 1144 Chapter 5 Removal and replacement

b. Continue to rotate the cover away from the printer to release two more tabs at the middle of the cover. Figure 5-1213 Release two tabs c. Slide the cover toward the rear of the printer to release two bosses at the front of the cover. Figure 5-1214 Slide the cover Removal and replacement: Paper feed drive assembly 1145

d. Remove the cover. Figure 5-1215 Remove the cover 10. Remove the top-rear cover (M527/E52545/M528/E52645). a. Remove one screw (callout 1), and then open the rear door. Figure 5-1216 Remove one screw 1 1146 Chapter 5 Removal and replacement

b. Rotate the exposed edge of the cover away from the printer (as shown). Figure 5-1217 Rotate the cover c. Continue to rotate the cover away from the printer to release one boss. Figure 5-1218 Release one boss Removal and replacement: Paper feed drive assembly 1147

d. Remove the cover. Figure 5-1219 Remove the cover 11. Remove the rear door assembly. a. At the rear of the printer, remove two screws (callout 1). Figure 5-1220 Remove two screws 1 1148 Chapter 5 Removal and replacement

b. Rotate the bottom edge of the assembly away from the printer to release it. Figure 5-1221 Rotate the bottom of the cover c. Pull down on the assembly to remove it. Figure 5-1222 Remove the cover 12. Remove the cartridge door assembly. NOTE: If the toner cartridge was not removed prior to servicing the printer, remove it now. a. Release the support shaft (callout 1), and then slide it toward the center of the door (callout 2). TIP: It might be easier to release the support shaft by using a small flat-blade screwdriver. Removal and replacement: Paper feed drive assembly 1149

Figure 5-1223 Release the support shaft 1 2 b. Remove the support shaft. Figure 5-1224 Remove the support shaft c. Release one tab (callout 1), and then slide the guide out of the printer (callout 2). CAUTION: Be careful, the tab (callout 1) can be easily broken. 1150 Chapter 5 Removal and replacement

Figure 5-1225 Remove the guide 2 1 d. Use a small flat-blade screwdriver to gently pry the white-plastic door arm (callout 1) off of the gray-plastic pressure-release link (callout 2) to release the door arm. Figure 5-1226 Release the pressure-release link 1 2 e. With the door in the partially closed position, slightly raise the left end (callout 1; raise it over the sheet-metal hinge boss), and then slide the door to the left (callout 2) to release the right hinge (callout 3). CAUTION: The door is still attached to the printer by the left-side retainer arm. Do not attempt to completely remove the door. Reinstallation tip: When reinstalling the door, there are two right-side hinge pins (not shown in callout 3). Keep a finger pressed on the left-side retainer arm so that it does not disengage from the door. Removal and replacement: Paper feed drive assembly 1151

Figure 5-1227 Release the door 3 1 2 f. Rotate the door away from the printer until the keyed pin on the left-side retainer arm can disengage from the door. Figure 5-1228 Release the left-side retainer arm 1152 Chapter 5 Removal and replacement

g. Remove the door. Figure 5-1229 Remove the door 13. Remove the right cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-1230 Locate the tabs and bosses Removal and replacement: Paper feed drive assembly 1153

b. Remove two screws (callout 1). Figure 5-1231 Remove two screws 1 c. At the rear of the printer, release one boss. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-1232 Release one boss d. At the bottom edge of the cover, release two bosses. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. 1154 Chapter 5 Removal and replacement

Figure 5-1233 Release two bosses e. Near the formatter, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Hold the bottom of the cover away from the printer, and then use a small flat-blade screw driver to easily release this tab. Figure 5-1234 Release one tab Removal and replacement: Paper feed drive assembly 1155

f. At the top rear edge of the cover, release one tab. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. TIP: Use a small flat-blade screw driver to easily release this tab. Figure 5-1235 Release one tab g. Rotate the rear of the cover away from the printer (callout 1), slide the cover toward the front of the printer (callout 2) to release two bosses (callout 3), and then remove the cover. CAUTION: It might be easier to remove the cover with the printer slightly hanging over the edge of the workspace. Do not let the printer tip and fall off of the workspace when removing the cover. Figure 5-1236 Remove the cover 2 1 3 1156 Chapter 5 Removal and replacement

14. Remove the left cover. IMPORTANT: The black-plastic cassette front and rear guides (located inside the tray cavity) can be easily dislodged. Do not lose the guides when the cover is removed or installed. NOTE: An MFP printer is shown in this section. However, the procedure is correct for all printer models. a. Before proceeding, take note of the location of the tabs and bosses on the inside of the cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs and bosses on the inside of the cover. Figure 5-1237 Locate of the tabs and bosses b. Remove one screw (callout 1). Figure 5-1238 Remove one screw 1 Removal and replacement: Paper feed drive assembly 1157

c. At the front of the printer, release one boss (callout 1), and then rotate the front edge of the cover away from the printer to release three tabs (callout 2). Figure 5-1239 Release one boss and three tabs 2 1 d. At the bottom edge of the cover, release two tabs. Figure 5-1240 Release two tabs e. Continue to rotate the front edge of the cover away from the printer (callout 1), and then slide the cover toward the rear of the printer (callout 2) to remove it. NOTE: If the cover is difficult to rotate or slide as shown below, make sure that the bottom tabs (released in the previous step) are completely disengaged from the chassis. 1158 Chapter 5 Removal and replacement

Figure 5-1241 Remove the cover 1 2 15. Remove the integrated-scanner assembly (ISA) (M527/E52545/M528/E52645). NOTE: The ISA includes the document feeder, sub-scanner assembly (SSA), and the scanner control board (SCB). a. Remove one screw (callout 1). IMPORTANT: This screw fastens the ISA ground plate to the printer chassis. Do not lose this screw. Figure 5-1242 Remove one screw 1 Removal and replacement: Paper feed drive assembly 1159

b. Disconnect three connectors (callout 1), pass the wire harnesses and cables through the opening in the cover (callout 2), and then release them from the retainers (callout 3). NOTE: One cable (callout 4) does not need to pass through the opening in the cover. TIP: It might be easier if the WiFi PCA and holder (not all models; callout 5) is detached from the printer, and then moved out of the way. Figure 5-1243 Disconnect cables and connectors 4 1 2 3 5 c. Disconnect three connectors (callout 1), and then remove six screws (callout 2). Figure 5-1244 Disconnect connectors and remove screws 1 2 1160 Chapter 5 Removal and replacement

d. Slightly slide the ISA towards the back of the printer to release it. Figure 5-1245 Slide the ISA towards the back of the printer e. Remove the ISA. Figure 5-1246 Remove the ISA 16. Remove the top cover (M527/E52545/M528/E52645). Removal and replacement: Paper feed drive assembly 1161

a. At the front right of the printer, remove one screw (callout 1), and then remove the sheet-metal plate (callout 2). Figure 5-1247 Remove one screw and the plate 1 2 b. At the top rear of the printer, remove three screws (callout 1). Figure 5-1248 Remove three screws 1 c. At the top

front of the printer, remove three screws (callout 1). NOTE: The screw (callout 2) on the lower-right portion of the cover is a different color than the other two screws, and is surrounded by sheet-metal. It might appear that this screw does not need to be removed. Remove this screw.  
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Figure 5-1249 Remove three screws 1 2 d. At the front left of the printer, release one boss. Figure 5-1250 Release one boss e. Rotate the front of the cover up and away from the printer. CAUTION: As the cover is rotated up, carefully pass the cables and wire harnesses the hole in the cover (callout 1). Removal and replacement: Paper feed drive assembly 1163

Figure 5-1251 Rotate the front of the cover up 1 f. Remove the cover. Figure 5-1252 Remove the cover 17. Remove the top cover (M501/M506/E50045/M507/E50145). The M506x/E50045x is shown in this procedure. However, the steps are correct for all M506/E50045/M507/E50145 models and the M501 printer. 1164 Chapter 5 Removal and replacement

a. Remove two screws (callout 1). Figure 5-1253 Remove two screws 1 b. M506/E50045/ only: Release one cable retainer from the chassis. TIP: Depress the tab on the retainer to release it. Figure 5-1254 Release one cable retainer Removal and replacement: Paper feed drive assembly 1165

c. M507/E50145 only: Disconnect two connectors. Figure 5-1255 Disconnect two connectors d. Release one boss. Figure 5-1256 Release one boss e. M506/E50045/ only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M506x/E50045x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. 1166 Chapter 5 Removal and replacement

Figure 5-1257 Remove the cover (M506/E50045) 1 f. M507/E50145only: Rotate the front of the cover up and away from the printer, and then remove the cover. CAUTION: M507x/E50145x only: As the cover is rotated up, carefully pass the cables the hole in the chassis (callout 1). IMPORTANT: If the top cover is being replaced, make sure to transfer the WiFi PCA from the discarded cover to the replacement cover. NOTE: LCD control panel models only: Disconnect the control panel before removing the cover. Figure 5-1258 Remove the cover (M507/E50145) 1 Removal and replacement: Paper feed drive assembly 1167

18. Remove the paper delivery tray (output bin). NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. a. Release two tabs (callout 1), and then slide the cover toward the front of the printer (callout 2). Figure 5-1259 Release two tabs 1 2 1 b. Remove the bin. Figure 5-1260 Remove the bin 19. Remove the formatter case. 1168 Chapter 5 Removal and replacement

a. M506/E50045/M507/E50145 only: Remove two screws (callout 1), and then remove the sheet metal plate (callout 2). NOTE: M507/E50145 only: The M507/E50145 sheet-metal plate includes a separate black plastic cable guide (not shown). Do not lose the guide when removing the plate. Figure 5-1261 Remove the sheet-metal plate 2 1 b. Carefully pass all cables and wire harnesses through the opening in the formatter case. NOTE: A SFP printer is shown in this section. However, the procedure is correct for all printer models. Figure 5-1262 Pass all cables and wire harnesses through the opening Removal and replacement: Paper feed drive assembly 1169

c. Remove one screw (callout 1), and then push down on one tab (callout 2) to release it. Figure 5-1263 Remove one screw 1 2 d. Remove the guide. Figure 5-1264 Remove the guide 1170 Chapter 5 Removal and replacement

e. Remove four screws (callout 1). Figure 5-1265 Remove four screws 1 f. Remove five screws (callout 1). Figure 5-1266 Remove five screws 1 Removal and replacement: Paper feed drive assembly 1171

g. Disconnect one connector (callout 1), and then move the guide out of the way (callout 2). Figure 5-1267 Disconnect one connector 2 1 h. Slightly lift up the formatter case (callout 1), rotate the top of the case away from the chassis (callout 2), and then lift up the case to release it (callout 3). CAUTION: Pass the FFC through the opening (callout 4) in the formatter while removing the case. Figure 5-1268 Release the formatter case 2 3 1 4 i. Remove the formatter case. NOTE: WiFi models only: If the formatter case is being replaced, make sure that the WiFi PCA and bracket (callout 1) are transferred to the replacement case. 1172 Chapter 5 Removal and replacement

Figure 5-1269 Remove the formatter case 1 20. Remove the LVPS. a. Remove one screw (callout 1), and then move the power-switch assembly (callout 2) out of the way. Figure 5-1270 Move the power-switch assembly 1 2 Removal and replacement: Paper feed drive assembly 1173

b. Disconnect one connector (callout 1), and then release the wire harness from the guide (callout 2). Figure 5-1271 Disconnect one connector 1 2 c. Disconnect two connectors (callout 1). TIP: Squeeze the locking lever on the lower connector to release it. Figure 5-1272 Disconnect two connectors 1 1174 Chapter 5 Removal and replacement

d. Remove two screws (callout 1). Figure 5-1273 Remove two screws 1 e. Release one tab (callout 1), and then rotate the top of the LVPS out and away from the printer (callout 2). Figure 5-1274 Release one tab 1 2 f. Slide the LVPS up and away from the printer to release it. Reinstallation tip: Push down on the LVPS to make sure that it is fully seated on the holder. Removal and replacement: Paper feed drive assembly 1175

Figure 5-1275 Release the LVPS g. Remove the LVPS. Figure 5-1276 Remove the LVPS 21. Remove the fan (FM2). 1176 Chapter 5 Removal and replacement

a. Disconnect one connector (callout 1), and then release the wire harness from the retainer (callout 2). Figure 5-1277 Disconnect one connector 1 2 b. Release two tabs (callout 1), and then rotate the top of the fan out and away from the printer (callout 2). Figure 5-1278 Release two tabs 2 1 c. Remove the fan. CAUTION: When reinstalled the fan must draw air into the printer. Make sure that the two arrows embossed on the right side of the fan body face downward and into the printer. Removal and replacement: Paper feed drive assembly 1177

Figure 5-1279 Remove the fan 22. Remove the fuser power supply (FPS). TIP: The FPS includes the optional paper feeder connector (Jetlink accessory tray connector) . a. Disconnect three connectors (callout 1). Figure 5-1280 Disconnect three connectors 1 1178 Chapter 5 Removal and replacement

b. Remove one screw (callout 1). Figure 5-1281 Remove one screw 1 c. Disconnect the tray accessory connector (circle callout) and release the wire harness from the guide. d. Release four tabs. TIP: It might be easier to release the upper two tabs by using a small flat-blade screw driver. Removal and replacement: Paper feed drive assembly 1179

Figure 5-1282 Release four tabs e. Rotate the left end of the assembly out and away from the printer (callout 1), slide the assembly to the left to release the right side alignment pins (callout 2), and then remove the assembly. Figure 5-1283 Remove the assembly 1 2 23. Remove internal cable and holder brackets. **IMPORTANT:** It is recommended to bring the printer back to the shop to perform the following removal and replacement repairs. **NOTE:** Lay the engine on its left side to make removal or the following parts easier. 1180 Chapter 5 Removal and replacement

a. Remove one (1) screw from the upper cable bracket. **NOTE:** The upper cable bracket assembly does not need to be completely removed, simply detached and then moved up out of the way and placed on the top of the printer. b. Remove the cable routing from the upper cable bracket. c. Lift up on the left side of the cable bracket (callout 1) and slide the bracket towards the front of the printer (callout 2) to remove the bracket. 1 2 Removal and replacement: Paper feed drive assembly 1181

d. Remove the fuser power supply bracket (small lower rear bracket) by pulling in the upper and left side of the bracket and lifting it from the printer. e. Remove the cables from the lower front cable bracket. 1182 Chapter 5 Removal and replacement

f. Remove three (3) screws (callout 1) from the bracket assembly. 1 g. Gently pry up the two (2) tabs on the lower right side of the bracket the remove the bracket from the printer. Removal and replacement: Paper feed drive assembly 1183

h. Remove two (2) screws (callout 1) from the metal M1 motor shield and remove the shield from the printer. 1 i. Remove three (2) screws (callout 1) and then remove the cable bracket located to the right of the main motor, M1. **NOTE:** Carefully lift off the bracket to ensure that the clutch on the back of the guide does not fall off when removing. 1 24. Remove the Paper feed drive assembly. **IMPORTANT:** It is recommended to bring the printer back to the shop to perform the following removal and replacement repairs. **NOTE:** Lay the engine on its left side to make removal or the following parts easier. 1184 Chapter 5 Removal and replacement

a. Remove six (6) black screws (callout 1). 1 b. Remove the paper feed drive assembly by gently lifting on the top side and remove it out from the printer. Reinstallation tip: When reinstalling, ensure the white drive gear is in the right orientation. You may need to adjust before reinstalling the feed drive assembly. Removal and replacement: Paper feed drive assembly 1185

25. Unpack the replacement assembly. a. Dispose of the defective part. **NOTE:** HP recommends responsible disposal of the defective part. <http://www8.hp.com/us/en/hp-information/environment/product-recycling.html> b. Unpack the replacement part from the packaging. **CAUTION:** Some parts are sensitive to electrostatic discharge (ESD). Always perform service work at an ESD-protected workstation or mat. If an ESD workstation or mat is not available, touch the sheet-metal chassis to provide a static ground before touching an ESD sensitive assembly. Protect the ESD-sensitive assemblies by placing them in ESD pouches when they are out of the printer. If the replacement part is a roller or pad, avoid touching the spongy part of it. Skin oils on the roller or pad can cause paper pickup and/or print-quality problems. HP recommends using disposable gloves when handling rollers or pads or washing your hands before touching rollers or pads. **IMPORTANT:** Make sure that all of the shipping materials (for example shipping tape) are removed from the replacement part prior to installation. c. To install an assembly, reverse the removal steps. **NOTE:** When applicable, special installation instructions are provided for an assembly at the end of the removal procedure. Always completely read the removal instructions and follow all special installation instructions. Removal and replacement: Paper delivery assembly Learn how to remove and replace the delivery assembly. **IMPORTANT:** It is recommended to bring the printer back to the shop to perform the following removal and replacement repairs. 1186 Chapter 5 Removal and replacement

Mean time to repair: 60 minutes Service level: Difficult

## Before performing service

### Turn the printer power off

■ Disconnect the power cable. **WARNING!** To avoid damage to the printer, turn the printer off, wait 30 seconds, and then remove the power cord before attempting to service the printer. Use the table below to identify the correct part number for your printer. To order the part, go to [www.hp.com/buy/parts](http://www.hp.com/buy/parts). Table 5-45 Part information

### Part number Part description

RM2-5684-000CN Simplex paper delivery assembly RM2-5683-000CN Duplex paper delivery assembly

### Required tools

- Small flat-blade screwdriver ● #2 Phillips screwdriver with a magnetic tip and a 152 mm (6 in) shaft length

### After performing service

### Turn the printer power on

■ Connect the power cable. ■ Use the power switch to turn the power on.

### Post service test

Make sure that the printer initializes to a Ready state. 1. Remove the control panel (M506x/E50045x/M507x/E50145x) Follow these steps to remove the control panel (M506x/E50045x/M507x/E50145x). Remove the control panel (M506x/E50045x/M507x/E50145x) 1187

1. Tilt the control panel to the fully raised position. Figure 5-1284 Tilt the control panel up 2. Use the small flat-blade screwdriver supplied in the kit to release the control-panel screws cover, and then remove the cover. **TIP:** The screwdriver has a reversible shaft. Simply pull the shaft out of the handle, and then reinstall it with the desired end facing out. Figure 5-1285 Remove the cover 1188 Chapter 5 Removal and

replacement

3. Use the small Phillips screwdriver supplied in the kit to remove two screws (callout 1). **IMPORTANT:** Store the screws in a secure place where they cannot fall down into the printer. Figure 5-1286 Remove two screws 1 4. Turn the control panel over (callout 1), to gain access to the bottom side. Figure 5-1287 Turn the control panel over 1 Remove the control panel (M506x/E50045x/M507x/E50145x) 1189

5. M506/E50045 only: Disconnect one HDMI cable (callout 1), one connector (callout 2), and then remove the control panel. Figure 5-1288 Remove the control panel (M506/E50045) 1 2 6. M507/E50154 only: If the formatter cover has not been removed, remove it now. Disconnect one flat-flexible cable (FFC). **CAUTION:** ESD sensitive part. **NOTE:** This connector is a zero-insertion force (ZIF) connector. Open the locking clip. The clip is easily dislodged and lost. Figure 5-1289 Disconnect one FFC (M507/E50145) 1190 Chapter 5 Removal and replacement

7. M507/E50145 only: Pass the FFC through the opening in the cover, and then remove the control panel. **CAUTION:** ESD sensitive part. Figure 5-1290 Remove the control panel (M507/E50145) 2. Remove the tray Follow these steps to remove the tray. 1. Pull the tray straight out of the printer until it stops. Figure 5-1291 Pull the tray out until it stops Remove the tray 1191

2. Lift the front of the tray, and then pull it out of the printer to remove it. **NOTE:** If the tray has Fixed Tray Guides installed and are still required, move the hardware to new tray. If necessary, order new kit: 3GY19-67901 Kit-Fixed Tray Guide. Only one kit required per engine. Figure 5-1292 Release and remove the tray 3. Remove the formatter cover Follow these steps to remove the formatter cover. ■ Do one of the following: ● M501: Remove one screw (callout 1), and then slide the cover toward the rear of the printer to remove it (callout 2). Figure 5-1293 Remove the formatter cover (M501) 2 1 1192 Chapter 5 Removal and replacement

● M506/E50045/M507/E50145/M527/E52545/M528/E52645: Slide the cover toward the rear of the printer to remove it. Figure 5-1294 Remove the formatter cover (M506/E50045/M507/E50145/M527/E52545/M528/

## E52645)

4. Remove the hard-disk drive (HDD models) Follow these steps to remove the hard-disk drive (HDD models). **CAUTION:** ESD sensitive part.

1. Release the locking connector (callout 1), and then pinch the retainer (callout 2) to release it. Figure 5-1295 Release the HDD 1 2 Remove the hard-disk drive (HDD models) 1193

2. Rotate the connector end of the HDD out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-1296 Remove the HDD 1 2 5. Remove the fax PCA (M527/E52545/M528/E52645 fax models) Follow these steps to remove the fax PCA (M527/E52545/M528/E52645 fax models). **CAUTION:** ESD-sensitive part. 1. Disconnect three connectors (callout 1). Figure 5-1297 Disconnect three connectors 1 1194 Chapter 5 Removal and replacement

2. Pinch the retainer to release it. Figure 5-1298 Release the fax PCA 3. Rotate the connector end of the fax PCA out and away from the formatter (callout 1), and then slide it as shown (callout 2) to remove it. Figure 5-1299 Remove the fax PCA 1 2 Remove the fax PCA (M527/E52545/M528/E52645 fax models) 1195

6. Remove the formatter Follow these steps to remove the formatter. **CAUTION:** Never remove and then install a replacement DCC and formatter at the same time. Doing so will cause the printer to become unstable. If a replacement DCC and formatter must be installed, first remove and install the formatter. Test operate the printer, and then if necessary install a replacement DCC. **IMPORTANT:** Take note of all empty connectors on the PCA before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessory items. 1. M501: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. Figure 5-1300 Remove the formatter (M501) 1 2. M506/E50045/M507/E50145/M527/E52545/M528/E52645: Disconnect all of the connectors and FFCs, remove four screws (callout 1), and then remove the formatter. **IMPORTANT:** Take note of all empty connectors on the formatter before disconnecting any wire harnesses or flat-flexible cables (FFCs). Some empty connectors might not be visible until other items—like the hard-disk drive—are removed, so pay close attention while removing the formatter and accessories items. 1196 Chapter 5 Removal and replacement

Figure 5-1301 Remove the formatter (M506/E50045/M507/E50145/M527/E52545/M528/E52645) 1 1 7. Remove the stapler door or stapler blank cover (M527/E52545/M528/E52645) Follow these steps to remove the stapler door or stapler blank cover (M527/E52545/M528/E52645). 1. Do one of the following: ● Stapler models: Open the stapler door. ● Non-stapler models: Use a small flat-blade screw driver to release two tabs (callout 1), and then rotate the cover away from the printer (callout 2). Figure 5-1302 Open the door or release the cover 2 1 2. Do one of the following: ● Stapler models: Use a small flat-blade screwdriver to release the upper hinge. ● Non-stapler models: Slide the cover toward the front of the printer to remove it. Remove the stapler door or stapler blank cover (M527/E52545/M528/E52645) 1197

Figure 5-1303 Release the upper hinge or remove the cover 3. Stapler models: Use a small flat-blade screwdriver to release the lower hinge. Figure 5-1304 Release the lower hinge 1198 Chapter 5 Removal and replacement

4. Stapler models: Remove the door. Figure 5-1305 Remove the door 8. Remove the stapler stationary (inner) cover (M527/E52545/M528/E52645) Follow these steps to remove the stapler stationary (inner) cover (M527/E52545/M528/E52645). **NOTE:** Stapler models only. 1. Before proceeding, take note of the locations of the upper tab (callout 1) and the lower tab (callout 2) on the stapler stationary (inner) cover. Reinstallation tip: When reinstalling the cover, take note of the location of the tabs. Figure 5-1306 Locate the tabs 1 2 Remove the stapler stationary (inner) cover (M527/E52545/M528/E52645) 1199

2. Use a long-shaft small flat-blade screwdriver to release the lower tab. Figure 5-1307 Release one tab 3. Use a long-shaft small flat-blade screwdriver to release the upper tab. Figure 5-1308 Release one tab 1200 Chapter 5 Removal and replacement

4. With the tabs disengaged the front of the cover is released. Rotate the front of the cover away from the printer (callout 1) to release the rear of the cover (callout 2), and then remove it. Figure 5-1309 Remove the cover 1 2 9. Remove the top-left cover (M527/E52545/M528/E52645) Follow these steps to remove the top-left cover (M527/E52545/M528/E52645). 1. Open the toner-cartridge door, release two tabs (callout 1),

and then rotate the rear edge of the cover away from the printer. Figure 5-1310 Release two tabs 1 Remove the top-left cover (M527/E52545/M528/E52645) 1201

2. Continue to rotate the cover away from the printer to release two more tabs at the middle of the cover. Figure 5-1311 Release two tabs 3. Slide the cover toward the rear of the printer to release two bosses at the front of the cover. Figure 5-1312 Slide the cover 1202 Chapter 5 Removal and replacement

4. Remove the cover. Figure 5-1313 Remove the cover 10. Remove the top-rear cover (M527/E52545/M528/E52645) Follow these steps to remove the top-rear cover (M527/E52545/M528/E52645). 1. Remove one screw (callout 1), and then open the rear door. Figure 5-1314 Remove one screw 1 Remove the top-rear cover (M527/E52545/M528/E52645) 1203

2. Rotate the exposed edge of the cover away from the printer (as shown). Figure 5-1315 Rotate the cover 3. Continue to rotate the cover away from the printer to release one boss. Figure 5-1316 Release one boss 1204 Chapter 5 Removal and replacement

4. Remove the cover. Figure 5-1317 Remove the cover 11. Remove the rear door assembly Follow these steps to remove the rear door assembly. 1. At the rear of the printer, remove two screws (callout 1). Figure 5-1318 Remove two screws 1 Remove the rear door assembly 1205

2. Rotate the bottom edge of the assembly away from the printer to release it. Figure 5-1319 Rotate the bottom of the cover 3. Pull down on the assembly to remove it. Figure 5-1320 Remove the cover 12. Remove the cartridge door assembly Follow these steps to remove the cartridge door assembly. NOTE: If the toner cartridge was not removed prior to servicing the printer, remove it now. 1. Release the support shaft (callout 1), and then slide it toward the center of the door (callout 2). TIP: It might be easier to release the support shaft by using a small flat-blade screwdriver. 1206 Chapter 5 Removal and replacement